

General Characteristics of Single-Phase Bi-Directional Meter “NEXY-M”

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AMENDMENT REGISTER

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1 Scope

This document has the scope to present the general characteristics of the single-phase bi-directional meter “NEXY-M” produced by E-distribuzione S.p.A. for the Static Meter Project for Low Voltage Customers.

The document deals with:

- general information and references about the manufacturer of the meter
- main performances and functionality's of the meter
- metrological characteristics and architecture description
- key elements and devices description
- design criteria and general layout of the product
- details on the electronic design
- layout of the case, measuring circuit and I/O terminals
- characteristics of the integrated cut-off device
- meter sealings and anti-tamper devices
- user and communication interfaces
- design options and equipments
- manufacturing process overview

This document is intended to be used as general introduction of the e-distribuzione static meters. It addresses all technical aspects those are considered relevant in a scenario of product certification requested by e-distribuzione to Institutes or Notified Bodies qualified for such an activity.

2 Applicability

This document is applicable to e-distribuzione single-phase bi-directional static meter (NEXY-M).

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3 Reference Standard

IEC EN 62052-11	Electricity metering equipment (AC) – General requirements, tests and test conditions – Part 11: Metering equipment
IEC EN 62053-21	Electricity metering equipment (AC) – Particular requirements – Part 21: Static meters for active energy (classes 1 and 2)
IEC EN 62053-23	Electricity metering equipment (AC) – Particular requirements – Part 23: Static meters for reactive energy (classes 2 and 3)
IEC 62053-61	Electricity metering equipment (a.c.) – Particular requirements – Part 61: Power consumption and voltage requirements
EN 50470-1	Electricity metering equipment (AC) – Part 1: General requirements, tests and test conditions – Metering equipment (class indexes A, B and C)
EN 50470-3	Electricity metering equipment (AC) – Part 3: Particular requirements – Static meters for active energy (class indexes A, B and C)
CLC EN 50065-1	Signalling on low-voltage electrical installations in the frequency range 3 kHz to 148,5 kHz Part 1: General requirements, frequency bands and electromagnetic disturbances.
IEC EN 60529	Degrees of protection provided by enclosures (IP Code)
EN 62056-21	Electricity metering – Data exchange for meter reading, tariff and load control – Part 21: Direct local data exchange
IEC EN 62058-11	Electricity metering equipment (AC) - Acceptance inspection - Part 11: General acceptance inspection methods
IEC EN 62058-31	Electricity metering equipment (AC) - Acceptance inspection - Part 31: Particular requirements for static meters for active energy (classes 0.2S, 0.5S, 1 and 2.
IEC EN 62052-31	Electricity metering equipment (AC) - General requirements, test and test conditions Part 31: Product safety requirements
EN 62059-31	Electricity metering equipment. Dependability. Accelerated reliability testing. Elevated temperature and humidity.
2014/32/UE	Directive of the European parliament and of the council of 26 February 2014 on the harmonization of the laws of the Member States relating to the making available on the market of measuring instruments
RD 244/2016	Real Decreto 244/2016, de 3 junio 2016 por el que se desarrolla la Ley 32/2014, de 22 de diciembre, de Metrología.
O.ICT/155/2020	ORDEN ICT/155/2020, de 7 de febrero, por la que se regula el control metrológico del Estado de determinados instrumentos de medida..
IEC/EN 62054-21	Electricity metering (a.c.) - Tariff and load control - Part 21: Particular requirements for time switches
Welmec 7.2:2019	Software Guide
Welmec 7.3	Reference Architectures based on Welmec Guide 7.2
Welmec 7.4	Exemplary Applications of Welmec Guide 7.2

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4 Acronyms and abbreviations

NEXY-M :	Single-Phase Bi-Directional Electronic Meter
LVM :	Low Voltage Manager
CERCO:	Low Voltage Concentrator.
DLC:	Distribution Line Carrier
HHU:	Hand Held Unit
LV:	Low Voltage
PCB:	Circuit board
LCD:	Liquid Circuit Display
SMMC	Monitoring and control measurement system

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5 Introduction

This document describes the design and construction of the electronic meter assembly and associated parts used in single-phase bi-directional DLC solid state electricity meter “NEXY-M”.

The design of the static meter has been based on a micro-controller that manages and coordinates all the activities related to the measuring and billing processes. It stores and runs the built-for-purpose application software downloaded into the micro-controller during the manufacturing (it is upgradeable). The energy measurement process is implemented by means of a dedicated IC. It is connected to the micro-controller so that energy data can be properly processed before the storing into non-volatile memory. The real time calendar is managed by the micro-controller itself. A chip on glass LCD display is installed into the meter in order to show measuring related data and personalized messages to the customer. The remote communication is performed by means of a DLC (integrated into the MCU) and RF modems whilst a ZVEI/Optical and Bluetooth interface have been implemented to transfer data to a local device.

The interaction with the customer is done by means of a simple push button located on the front side of the meter. Finally, the meter is equipped by a cut-off device that allows the operator to disconnect the customer and, in case, allow the reconnection.

The design of the meter has been developed in order to be compliant with all relevant international standard normative defined in the electricity metering field (a detailed list of international normative conformities will be given in the next paragraphs).

The electronics have been designed using solid-state surface mount technology wherever possible. The choice of components has been based on cost and availability. e-distribuzione has designed the mechanical layout of the PCB in partnership with the meter case designers to ensure that the complete product can be transferred directly from the prototype stage into production. The component layout has been carried out using best practices for high volume manufacturing in order to reduce assembly time and cost to a minimum.

Careful consideration has also been given to the position and number of test points that are required for Testing Circuit during the manufacturing stage.

The general construction of NEXY-M is based on one multilayer PCBs (main PCB and 169 MHz and Bluetooth), one double layer PCB (power supply) and LCD display. The PCBs materials are standard as is the population and soldering requirements. Both the surface mount components and the through hole components are assembled on top side of the main PCB and on the bottom side of the Power supply PCB.

The PCBs are assembled into a plastic meter case as long as the cut-off device, terminals, connectors and sealings. The plastic of the meter case has been chosen in order to maintain its characteristics (e.g. strength, insulation, etc) in a wide temperature range.

The meter case is also equipped by a transparent window in order to make the LCD display visible by the outside. A plastic terminal cover is used to protect the input and output terminals of the meter.

The nameplate is written by means of a laser-burning process so it results indelible.

When the meter is closed and installed on the DIN socket, accessing to PCB, electronic devices, shunts and power supply is absolutely forbidden.

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6 Main performances of the meter

The meter is designed to measure both active and reactive energy. Active energy is measured with a maximum resolution of 1 Wh whilst reactive energy with a maximum resolution of 1 varh.

Active energy measurement accuracy is compliant with "Class 1" defined into IEC/CEI EN 62053-21. It is also compliant with accuracy "CLASS B" defined into EN 50470-3.

Reactive energy measurement accuracy is compliant with "Class 2" defined into IEC/CEI EN 62053-23.

The **pulse emitter**, which can be used for either active energy or reactive energy, is located on the meter case can have two different resolution (factory configurable):

- or 4000 pulses per kWh and 4000 pulses per kvarh resolution.
- or 1000 pulses per kWh and 1000 pulses per kvarh resolution (*only in a second stage of the project*).

The meter is designed to be a bi-directional meter with four quadrants registers. It means that positive active energy, negative active energy, positive inductive reactive energy, positive capacitive reactive energy, negative inductive reactive energy and negative capacitive reactive energy are processed and stored in separate registers.

The meter has been designed in order to properly run in the following **temperature range**: $[-40^{\circ}\text{C} \div +70^{\circ}\text{C}]$.

The operating **frequency range** supported by the meter is: [50 Hz] or [60 Hz] (factory configurable).

The meter has a standard **reference voltage** equal to 230 V or 127V...230V (factory configurable) and **reference currents**: $I_{min} = 0.25\text{A}$, $I_{ref} = 5\text{A}$ and $I_{max} = 60\text{A}$ or 100A (*factory configurable – 100A only in a second stage of the project, with a different cut-off device*).

The internal consumption of the meter is inferior to 1.5 W when the DLC modem is in receiving mode (2 W when the DLC modem is in transmission mode) in conformity with the requirements of applicable standards (EN 50470-3, IEC 62053-21, IEC 62053-23 and IEC 62053-61)

The DLC modem has been designed to be compliant with CENELEC EN 50065-1.

The plastic case of the meter assures a degree of protection equals to **IP54** (according to CEI EN 50470-1) for the metrology compartment. The front clamp assures a degree of protection equals to IP20 when cables and terminal cover are installed.

The meter is intended for **indoor installation**.

The battery has been designed in order to assure the operation of RTC for 15 years without line voltage and it is not removable.

The meter has been designed to be compliant to protective "class II".

The meter has been designed to be compliant to IEC 62052-31 safety standard.

The meter has been designed and produced in order to assure its full functionalities all over the life-time of the meter (15 years).

The meter has been designed as a RoHS compliant product.

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7 Meter's main functionalities

NEXY-M meter has been designed in order to support two different communication protocol (factory configurable):

- Meters and More;
- DLMS

This feature is guaranteed thanks the implementation of two different “families of software”: one that supports Meters and More communication protocol and one DLMS. The two families of software are not interoperable: it means that if a meter is produced with a Meters and More software, it cannot be update with a DLMS software.

Even if, the communication protocols are different, the functionalities supported are the same.

The global functionalities, typical of the meter, are listed below.

- It support software separation between legally relevant and not legally relevant software;
- It operates in remote controlled mode and in stand-alone mode
- Active energy and Active power measurement, positive and negative
- RMS current and RMS voltage measurement
- Reactive energy and Reactive power measurement in all four quadrants
- Line of communication inversion immunity
- Load profile recording, for active and reactive energy in all four quadrants. The integration period for load profile recording is programmable and it can be chosen between the following values: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30 or 60 minutes (with an integration period of 15 minutes, it is possible to store information for more than 45 days).
- Management of customer contracts for billing purposes
- Management of 6 weekly tariff profiles based on 6 different types of tariff and 8 daily time intervals (one set for each contract)
- Management of a seasonal tariff program with up to 10 periods
- Management of programmable public holidays
- Management of 7 billing periods (current and 6 previous). For each billing period is stored, measurement data, power maximum demand and time stamp of the billing period closure.
- Management of daily closure
- **DLC/RF communication with LV concentrator (CERCO1 or LVM) supporting access control**
- Customer available power threshold is remotely programmable
- Display of consumption data and service communications (displayed data are those measured by the meter).

The meter allows two display modes: automatic and manual. The default mode is the manual one which cyclically shows the basic information for customer without the need to interact with the meter. Through the manual mode it is possible to activate submenus (by pressing the push button) in which additional information is shown to support technical and qualified personnel (date, time, software version...).

- Remote programming and initialization of tariff system and contractual parameters
- Remote synchronisation of Clock/calendar. Synchronisation doesn't affect historical data and it can be

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managed in order to prevent difference of more than a programmable threshold (e.g. 3 minutes) between the official time and the time of the meter.

- Clock/calendar of the meter is configurable in a flexible way: it can be referred to local time (according to national standard for quality of service) without variation during of the year or implementing automatically the DST time activation. The time adjustments that can be necessary for other purposes are made by the respective systems of each Distribution Company, without affecting the time settings of the SMMC components.
- Remote supply disconnection and enabling of manual/automatic connection by means of a cut-off device
- Zload functionality;
- Storage of configuration data and metrological information even without power supply for the entire lifetime of the meter (storage in non-volatile memory)
- Self diagnostic of the main elements and functional blocks
- Detection and recording of case openings and/or SW modifications
- Detection strong external magnetic fields by magnetometer installed on PCB;
- Optical pulse output device used for testing the meter (active or reactive energy)
- Non-interfering harmonization with other communication systems on LV network
- **ZVEI optical port communications compatible with CEI EN 62056-21, required for local communications**
- Load modulation management (load shedding)
- Management of firmware downloading procedure;
- Management of traceability of Firmware Download;
- Data and firmware integrity verification in compliance to Welmec guide 7.2 requirements;
- RF spontaneous management
- Event Log;
- Monitoring of microcontroller functionality;
- Voltage variation management: the meter is able to detect and store Low Voltage Variation respect to the nominal value. In particular the meter is able to detect (and store information) when the voltage goes above 10% (configurable parameter) or below 15% (configurable parameter) the nominal mains voltage, according to the requirements of the applicable standards of quality of service;
- Voltage Interruption management. If a voltage interruption occurs, the meter is able to save – before switching off – all legally relevant information and those relating to the state of the relè (if it is closed or open and if a power limitation is active). In this way, when the power supply is restored, the meter can resume the operation mode it had before the interruption (including communication).

Furthermore, the meter support bi-directional communication with SMMC (through data Concentrator or Smartphone/PDA) with four communication technologies:

- Remote:
 - Wireless @ 169 MHz;
 - Power Line;
- Local

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- Bluetooth (only support, not already implemented);
- Optical port.

Local and remote communication is protected through an authentication and encryption mechanism with secret keys (different for each counter) of 16 bytes. Each access is tracked in an internal log. Access attempts with wrong keys are also tracked with the activation of an alarm when the number of attempts exceeds a certain (programmable) threshold.

Components of the SMMC also comply with applicable regulations and defined in section 3 *Reference Standard*.

Optical, Bluetooth and Power Line channels support only point to point communication, while RF can support also (depending on the management of the secret keys) point to multipoint communication.

The SMMC in its local access is able to:

- Access the data stored in the meter's memory.
- Download the data stored in the meter's memory.
- Access and modify configurable parameters of the meter
- Access a component operation diagnosis in order to resolve anomalies and – if lost - reestablish remote communication with the Management and Operation System and / or Concentrator Unit.
- Access the functions of the meter

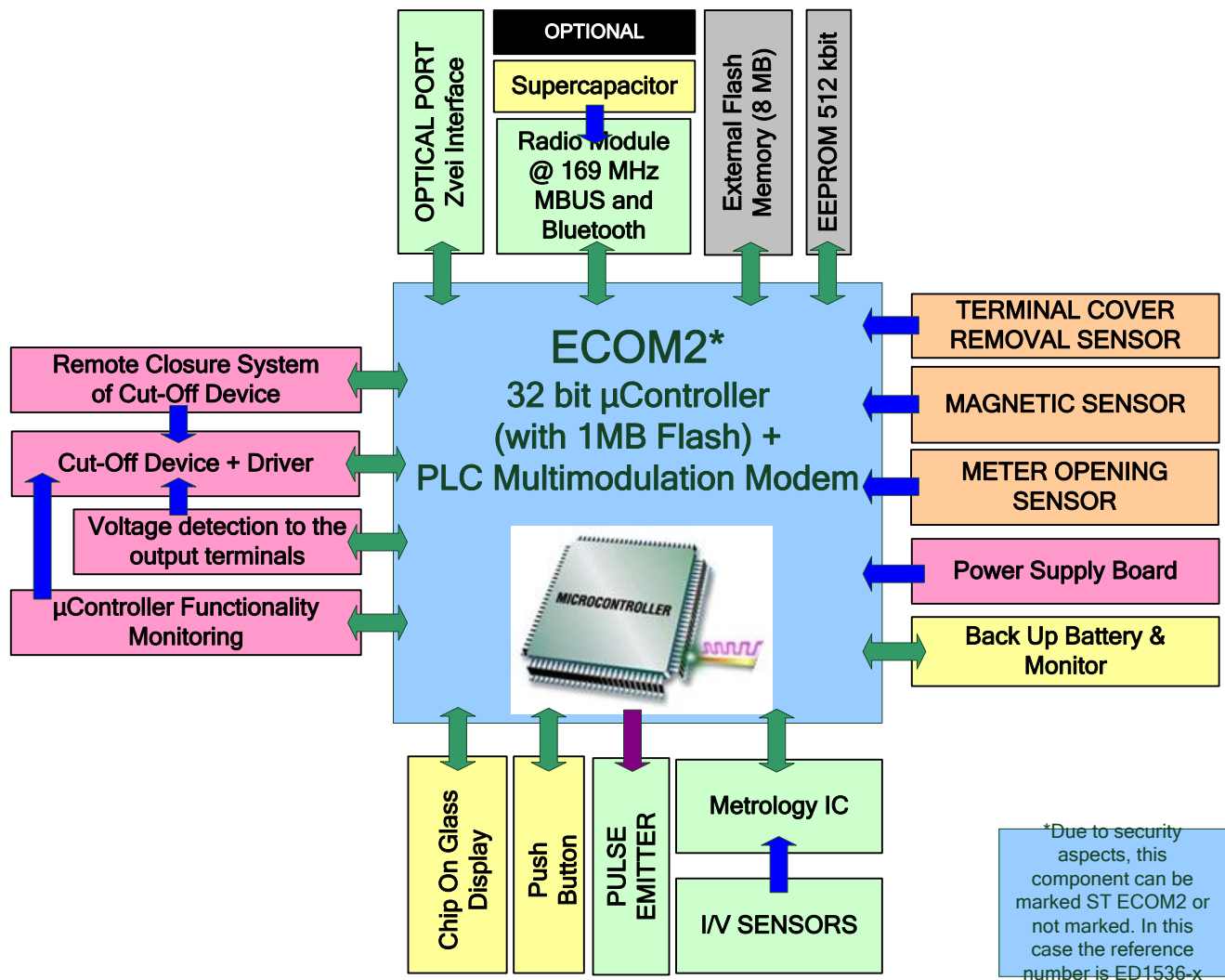
The meter implements (through one of the previously described channels) a two-way communication with the Management and Operation System. Thanks to this, it is possible to perform (locally or remote) the connection, disconnection and power limitation operations. It is also possible to know at any time the status of the connection and disconnection device and the power limitation.

All the functionalities listed above (included those related to local and remote communication) will be thoroughly described in a separate document that addresses software aspects of the meter.

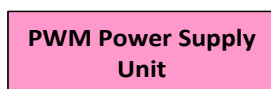
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8 Meter block diagram

Block diagram below demonstrates the centralized architectural approach that has been used for the single-phase meter



8.1 Detailed description of the Block Diagram



The electronics are powered by a two level voltage (+3,3Vdc, +3,6Vdc, +5Vdc and +10Vdc) with output protection. It has been designed as a “Pulse Width Modulation” mode Power Supply and capable of withstanding 115Vac and 440 Vac continuous overvoltage condition.

ECOM2
32 bit μ Controller
(with 1MB Flash) +
PLC Multimodulation Modem



ECOM2 is a device that integrates a power line communication modem and a high-performance application core.

The integrated ARM 32-bit Microcontroller with up to 1 MB of FLASH memory is used to integrate all of the meter functions including: billing data management, user interfaces, LCD management, load profiling, RTC operation, cut-off control, tamper detection, etc.

The PLC modem and coupling circuit are used to handle the transmission of data along the LV power lines by difference mode. ECOM supports different kind of modulation and in a different communication bands.

Due to security aspects, this component can be marked ST ECOM2 or not marked. In this case the reference number is ED1536-x.

**Chip On Glass
Display**

A custom STNx16 character Dot Matrix Chip-on-Glass Display provides visual indication of the meter status and parameters. The display also allows for variations in language and customized graphics. STN glass is used to provide maximum viewing angle and temperature ratings.

**Push
Button**

A push button allows messages and meter parameters to be cycled on the LCD display.

**PULSE
EMITTER**

The pulse emitter for the Energy consumption indicator emits, as standard, Ultra Bright RED LEDs with a meter constant of 4.000 pulses/kWh, kvarh or 1.000 pulses/kWh, kvarh

EEPROM

An EEPROM (512 kbit) is used to store all of the meters parameters profiling data and energy registers. This device has its own data bus connected to the MCU.

**EXTERNAL FLASH
MEMORY**

A Flash memory (8Mbyte) is installed into the meter in order to integrate the quantity of flash memory of the micro-controller.

**OPTICAL PORT
Zvei Interface**

An EN 62056-21 infrared optical port allows communication between a HHU and the meter. The port also allows the modification of all meter parameters by means of specific commands, including program code update (download of a new software code). This communications port uses the same security and protocols as the DLC modem and is managed directly by the DLC modem.

**TERMINAL COVER
REMOVAL SENSOR**

The PCB is equipped with one main anti-fraud measures. It is represented by a mechanical tamper detection switch that can be triggered by removal of the meter terminal cover and can be reset remotely by the PLC system or optical port.

MAGNETIC SENSOR

Magnetic sensor installed into the PCB for strong external magnetic fields detection.

**METER OPENING
SENSOR**

The Meter is provided with a micro-switch installed in the PCB, able to detect the removal of the cover of the meter.

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METROLOGY IC	Active and Reactive power measurement is carried out by the measurement IC ADE 7953. The corresponding accuracy classes are: CLASS “B” for active energy (according to EN 50470-3) and CLASS 2 for reactive energy (according to EN 62053-23) as per ENEL requirements.
I/V SENSOR	Active and Reactive power measurement is carried out by collecting I/V signals by the LV phase (with shunt) and neutral (with CT). These signals are provide to A/D converters integrated into the MCU.
BACK UP BATTERY + MONITOR	A backup battery is used in order to power the RTC and anti-tamper device when the main power is not available. A battery monitor IC is used in order to evaluate the status of the internal battery.
Cut-Off Device + Driver	The MCU is able to electronically command the cut-off device integrated into the meter by means of a driver circuit. The cut-off device allows to be opened by remote.
Remote Closure System of Cut-Off Device	The cut-off device allows to be enabled for closure and closed by remote (or through a communication protocol command or through “Z-Load” Functionality).
Voltage detection to the output terminals	The meter is able, thanks to dedicated components, to detect the voltage at the relay output terminals to detect any anomalies (voltage present with relay open).
μController Functionality Monitoring	The correct operation of the microcontroller can be permanently monitored through a circuit that, if it detects possible tampering actions, it can activate an alarm and take action on the circuit breaker.

There are some components that can be considered as optional into Single-Phase meter “NEXY-M”. These components are not related to metrological functionality, but to security or not legally relevant functionalities. Depending on the needs, during the assembling process, these components can be mounted on the PCB or not. In any case, the design of the meter is unique: in fact, the PCBs are designed in order to work with or without these components. Also the software of the meter is unique and can work independently from hardware configuration (only for optional components).

OPTIONAL Radio Module @ 169 MHz WMBUS and Bluetooth and Supercap	The meter can be equipped with an optional radio module (with both 169 MHz WMBUS and Bluetooth communication capabilities), Antenna and circuitry. Thanks to this optional component the meter can use the RF (at 169 MHz Frequency Band) as remote back up communication channel and Bluetooth as local back-up communication channel.
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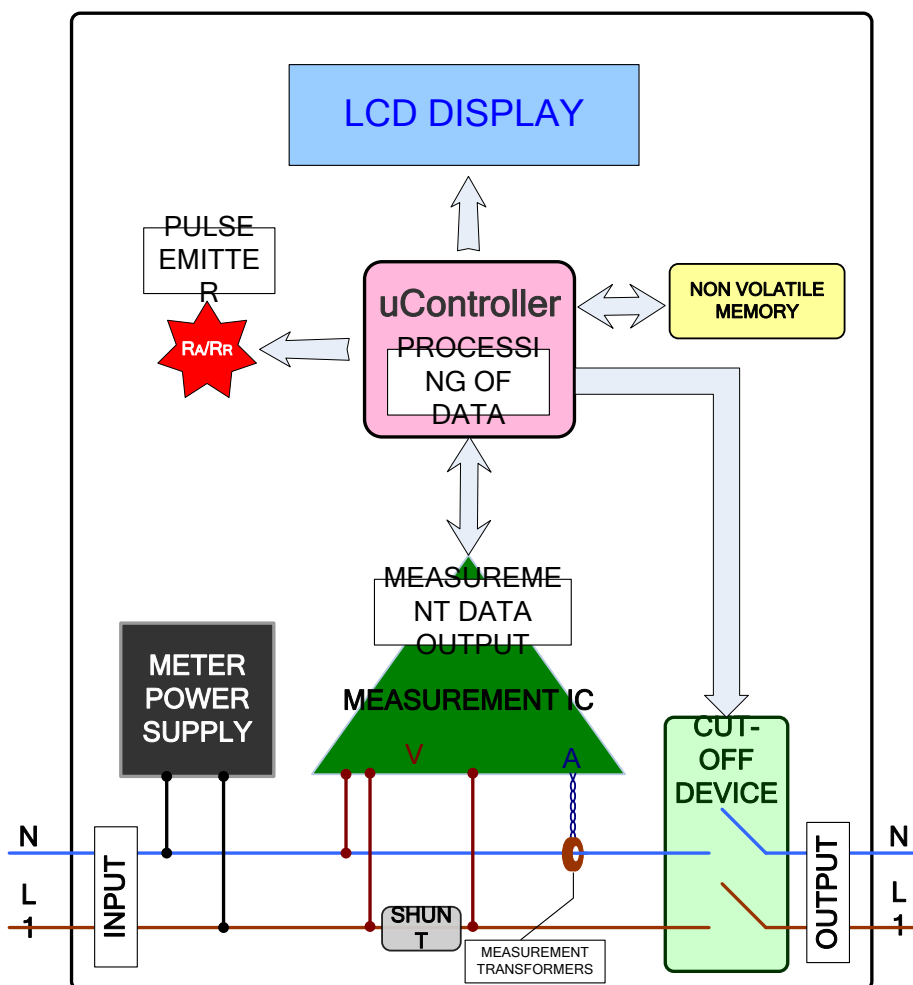
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9 Metrological characteristics

The measurement system of the NEXY-M meter is composed by three different key elements:

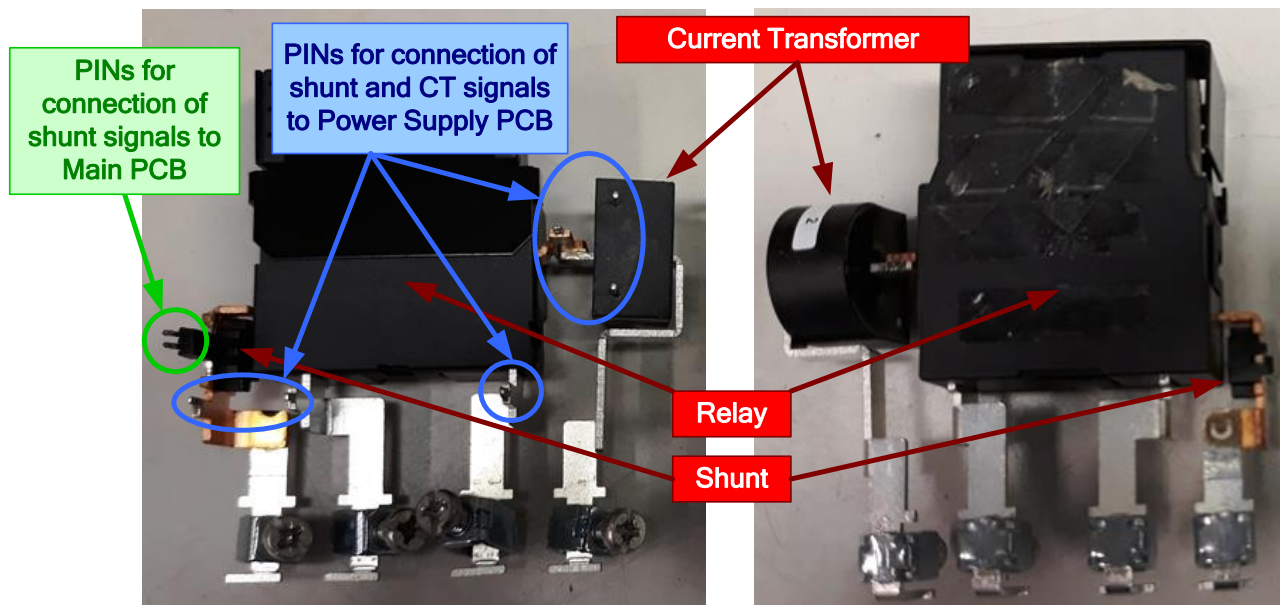
1. Measurement IC
2. Shunt
3. Measurement Transformer

A principle outline of the energy and power measurement functionality provided by the Single-Phase meter is shown below:

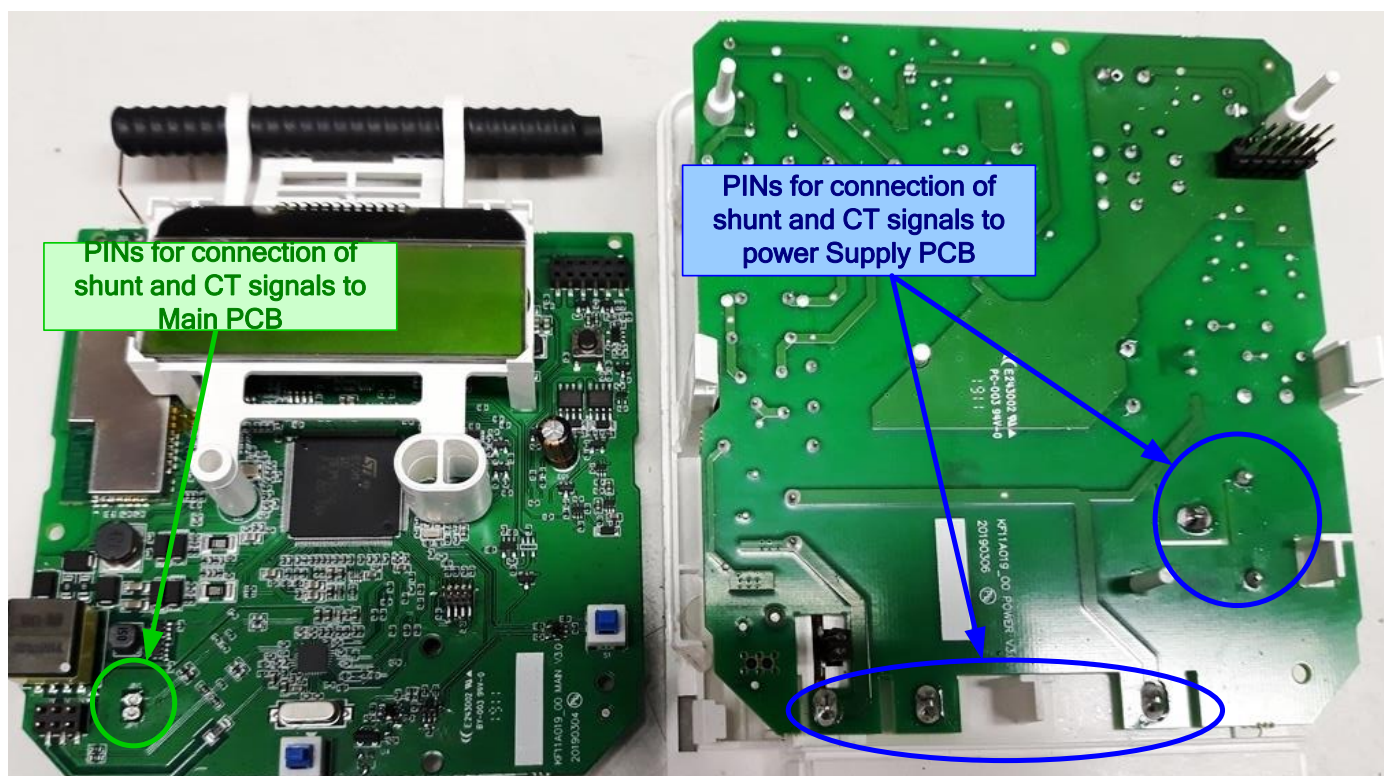


The power supply of the meter is connected before the Measurement IC so that the meter does not measure self-consumption.

Shunt and Current transformer are assembled directly on the terminals of the relay and soldered on the PCBs (shunt: 2 pins to the main board and three on power supply board, CT all pins to Power supply board) without additional cables.



Front and rear side of relay block with details of shunt and current transformer pins for connection to the PCBs



Details of soldering point of V/I signal of CT and shunt on the PCBs

The **Shunt** is serial connected on the phase line. This element provides a conversion I/V of the current flowing on the **phase line** absorbed by the load connected to the OUTPUT. This signal is then connected by the Measurement IC.

The **Current transformer** is connected on the neutral line. This element provides a conversion I/V of the current flowing on the **neutral** absorbed by the load connected to the OUTPUT. This signal is then collected by the Measurement IC.

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The Measurement IC collects the required signals (line voltage and voltage signal from **shunt**) as described in the schematic. These signals are used to evaluate **power and energy consumption data** that is transferred to the uController for the subsequent processing (e.g. tariff management).

The Measurement IC collects also the required signals (line voltages and current signals) from **measurement transformer** as described in the schematic. These signals are used to evaluate the **current** flowing in the **neutral** to detect a possible fraud attempt comparing this measurement with the one performed using the signal taken from the shunt. So, this measure is not used to evaluate energy consumption data or for other subsequent processing.

The uController provides all the functionalities required for the correct calibration of the Measurement IC.

The uController also gives support to:

- store measurement data into non-volatile memory
- control the pulse emitter for reactive energy
- control the LCD display
- control the trip command to disconnect and/or allow reconnection of the output switch

9.1 Measurement IC description (ADE 7953)

e-distribuzione NEXY-M meters are equipped with a measurement-dedicated IC (Integrated Circuit) provisioned by Analog Device.

The specific IC model for single phase meter is: “ADE 7953 – Single-Phase Multifunction Energy Metering IC with Neutral Current Measurement”.

For any details about the functions provided by the measurement IC, please refer to the related Datasheet (http://www.analog.com/static/imported-files/data_sheets/ADE7953.pdf).

Active Power Calculation performed by ADE7953

Power is defined as the rate of energy flow from source to load. It is defined as the product of the voltage and current waveforms. The resulting waveform is called the instantaneous power signal and is equal to the rate of energy flow at every instant of time. The unit of power is the watt or joules/sec. The following equation gives an expression for the instantaneous power signal in an AC system.

$$v(t) = \sqrt{2} \cdot V \sin(\omega t)$$

$$i(t) = \sqrt{2} \cdot I \sin(\omega t)$$

Where:

V is the RMS voltage.

I is the RMS current.

$$p(t) = v(t) \times i(t)$$

$$p(t) = VI - VI \cos(2\omega t)$$

The average power over an integral number of line cycles (n) is given by the following expression:

$$P = \frac{1}{nT} \int_0^{nT} P(t) dt = VI$$

Where:

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T is the line cycle period.

P is referred to as the active or real power.

The active power is equal to the DC component of the instantaneous power signal $p(t)$ in the previous equation.

The active power is therefore equal to VI . This is the relationship used to calculate active power in the ADE7953.

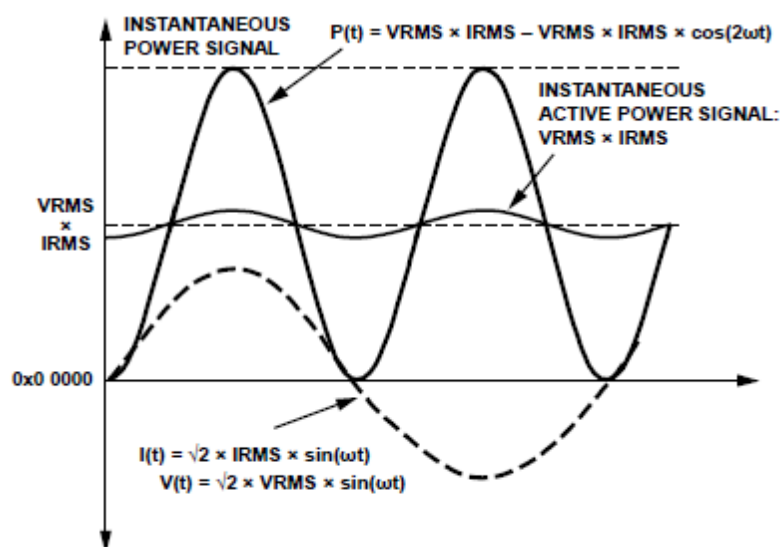
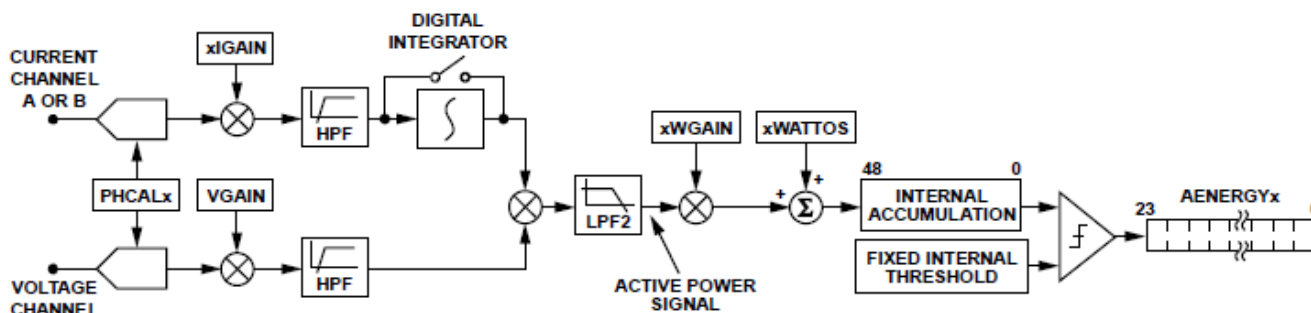


Fig. 1 - Active Power Calculation

The signal chain for the active power and energy calculation in ADE7953 is shown in the following picture.



The instantaneous power signal $p(t)$ is generated by multiplying the current and voltage signals. The DC component of the instantaneous power signal is then extracted by LPF2 (low-pass filter) to obtain the average active power information.

The ADE7953 computes the active power simultaneously on two channel (Current Channel A and Current Channel B) and stores the resulting measurements in the AWATT and BWATT registers, respectively. With full-scale inputs, the expected reading in the AWATT and BWATT registers is approximately 4862401 LSBs (decimal).

The active power measurements are taken over a bandwidth of 1.23 kHz and include the effects of any harmonics within that range. The active power registers are updated at a rate of 6.99 kHz and can be read using the waveform sampling mode.

The active power measurement in the ADE7953 is a signed calculation. If the phase differential between the current and voltage waveforms is more than 90° , the power is negative. Negative power indicates that energy is being injected back into the grid. The ACCMODE register includes two sign indication bits that show the sign of the active power of Current Channel A (APSIGN_A) and Current Channel B (APSIGN_B).

Active Energy Calculation performed by ADE7953

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The ADE7953 achieves the integration of the active power signal in two stages. In the first stage, the active power signals are accumulated in an internal 48-bit register every 143 μs (6.99 kHz) until an internal fixed threshold is reached. When this threshold is reached, a pulse is generated and is accumulated in 24-bit, user-accessible accumulation registers. The internal threshold results in a maximum accumulation rate of approximately 206.9 kHz with full-scale inputs. This process occurs simultaneously on Current Channel A and Current Channel B, and the resulting readings can be read in the 24-bit AENERGYA and AENERGYB registers. Both stages of the accumulation are signed and, therefore, negative energy is subtracted from positive energy.

This discrete time accumulation, or summation, is equivalent to integration in continuous time. The following Equation expresses this relationship:

$$E = \int p(t)dt = \lim_{T \rightarrow 0} \left\{ \sum_{n=1}^{\infty} P(nT) \times T \right\}$$

Where:

n is the discrete time-sampled number.

T is the sample period.

Reactive Power Calculation performed by ADE7953

Reactive power is defined as the product of the voltage and current waveforms when one of these signals is phase-shifted by 90° . The resulting waveform is called the instantaneous reactive power signal.

The two equations below represent the instantaneous voltage and current of an AC system:

$$v(t) = \sqrt{2} \cdot V \sin(\omega t - \theta)$$

$$i(t) = \sqrt{2} \cdot I \sin(\omega t)$$

The current shifted by $+90^\circ$ is described by the following expression:

$$i'(t) = \sqrt{2} \cdot I \sin\left(\omega t + \frac{\pi}{2}\right)$$

Where:

θ is the phase difference between the voltage and current channel

V is the RMS voltage

I is the RMS current

The following equations give an expression for the instantaneous reactive power signal in an AC system when the phase of the current channel is shifted by $+90^\circ$

$$q(t) = v(t) \times i'(t) = VI \times \sin \theta + VI \times \sin(2\omega t + \theta)$$

The average power over an integral number of line cycles (n) is given in the following equation:

$$Q = \frac{1}{nT} \int_0^{nT} q(t)dt = VI \times \sin(\theta)$$

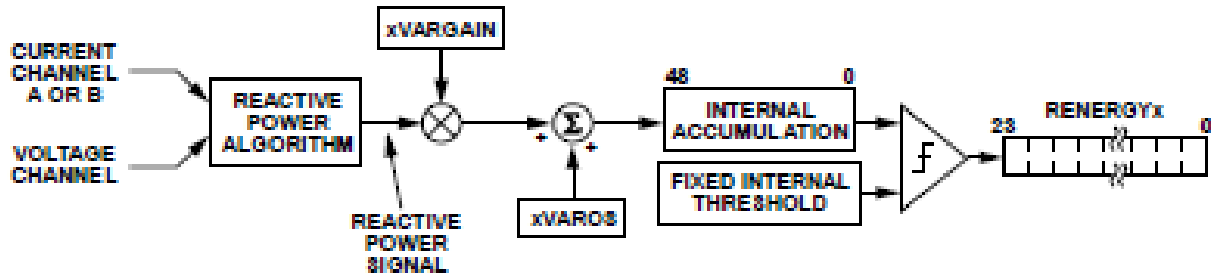
Where:

T is the line cycle period

Q is referred to as the reactive power

The reactive power is equal to the DC component of the instantaneous reactive power signal $q(t)$. This relationship is used to calculate reactive power in the ADE7953. The signal chain for the reactive power and energy calculations in the ADE7953 is shown in the following figure.

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The instantaneous reactive power signal $q(t)$ is generated by multiplying the current signal and the voltage signal. Simultaneous calculations are performed using Current Channel A and Current Channel B. The multiplication is performed over the full 1.23 kHz bandwidth and results in a reactive power measurement that includes all harmonics included in this range.

The ADE7953 computes the reactive power simultaneously on Current Channel A and Current Channel B and stores the resulting measurements in the AVAR and BVAR registers, respectively. With full-scale inputs, the expected reading in the AVAR and BVAR registers is approximately 4862401 LSBs (decimal).

The reactive power registers are updated at a rate of 6.99 kHz and can be read using the waveform sampling mode.

The reactive power measurement in the ADE7953 is a signed calculation. If the current waveform is leading the voltage wave-form, the reactive power is negative. Negative reactive power indicates a capacitive load. If the current waveform is lagging the voltage waveform, the reactive power is positive. Positive reactive power indicates an inductive load. The ACCMODE register includes two sign indication bits that show the sign of the reactive power of Current Channel A (VARSIGN_A) and Current Channel B (VARSIGN_B).

Reactive Energy Calculation performed by ADE7953

The ADE7953 achieves the integration of the reactive power signal in two stages. In the first stage, the reactive power signals are accumulated in an internal 48-bit register every 143 μ s (6.99 kHz) until an internal fixed threshold is reached. When this threshold is reached, a pulse is generated and is accumulated in 24-bit, user-accessible accumulation registers. The internal threshold results in a maximum accumulation rate of approximately 206.9 kHz with full-scale inputs. This process occurs simultaneously on Current Channel A and Current Channel B, and the resulting readings can be read in the 24-bit REENERGYA and REENERGYB registers. Both stages of the accumulation are signed and, therefore, negative energy is subtracted from positive energy.

This discrete time accumulation or summation is equivalent to integration in continuous time.

$$\text{Reactive Energy} = \int q(t)dt = \lim_{T \rightarrow 0} \left\{ \sum_{n=1}^{\infty} q(nT) \times T \right\}$$

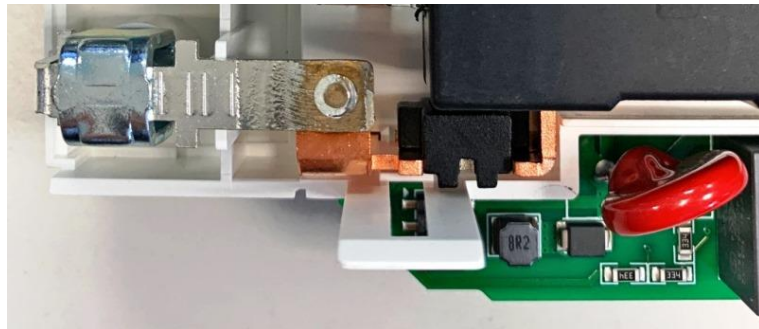
Where:

- $q(t)$ is the instantaneous reactive power
- n is the discrete time sample number
- T is the sample period

9.2 Shunt description

The shunt is a key component in the design of the power measurement circuits. In single phase meter “NEXY-M”, the shunt is perform measurement on the phase line.

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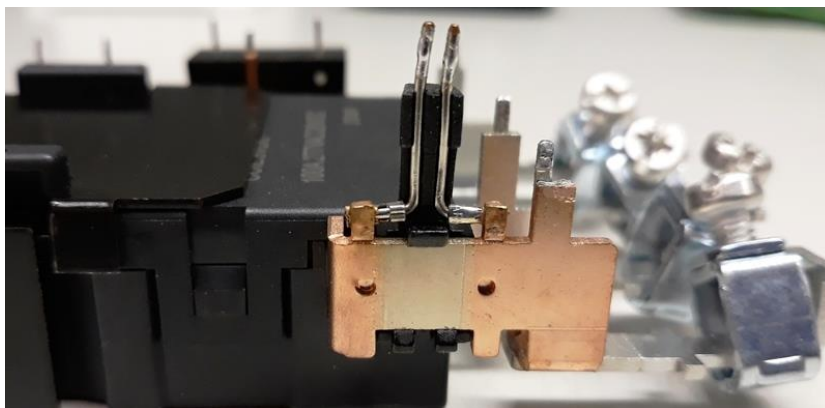


The choice of material and size of shunt determine the characteristics of the circuit with temperature. It is therefore paramount that the assembler manufactures the shunt in accordance with e-distribuzione's specification to ensure conformity to the e-distribuzione requirements. The base material for the manufacture of the shunt is manganine (the composition is CU-86 & Mn-12 & Ni-2). The characteristics of the shunt element are described below:

- Shunt resistance across measuring contact = $280 \mu\Omega$ (if the $I_{max} = 60A$) or $184 \mu\Omega$ (if the $I_{max} = 100A$)
- Max resistance tolerance = $\pm 15\%$
- Thermal stability $\leq 0,6\%$ ($0^\circ C \div 150^\circ C$)

Regular checks on the welded shunt assemblies is carried out as part of quality control with mechanical strength test of the shunt output.

The shunt is directly connected to the relay phase line input and it is soldered to the PCB in order to avoid other cables connection.



Detail of shunt

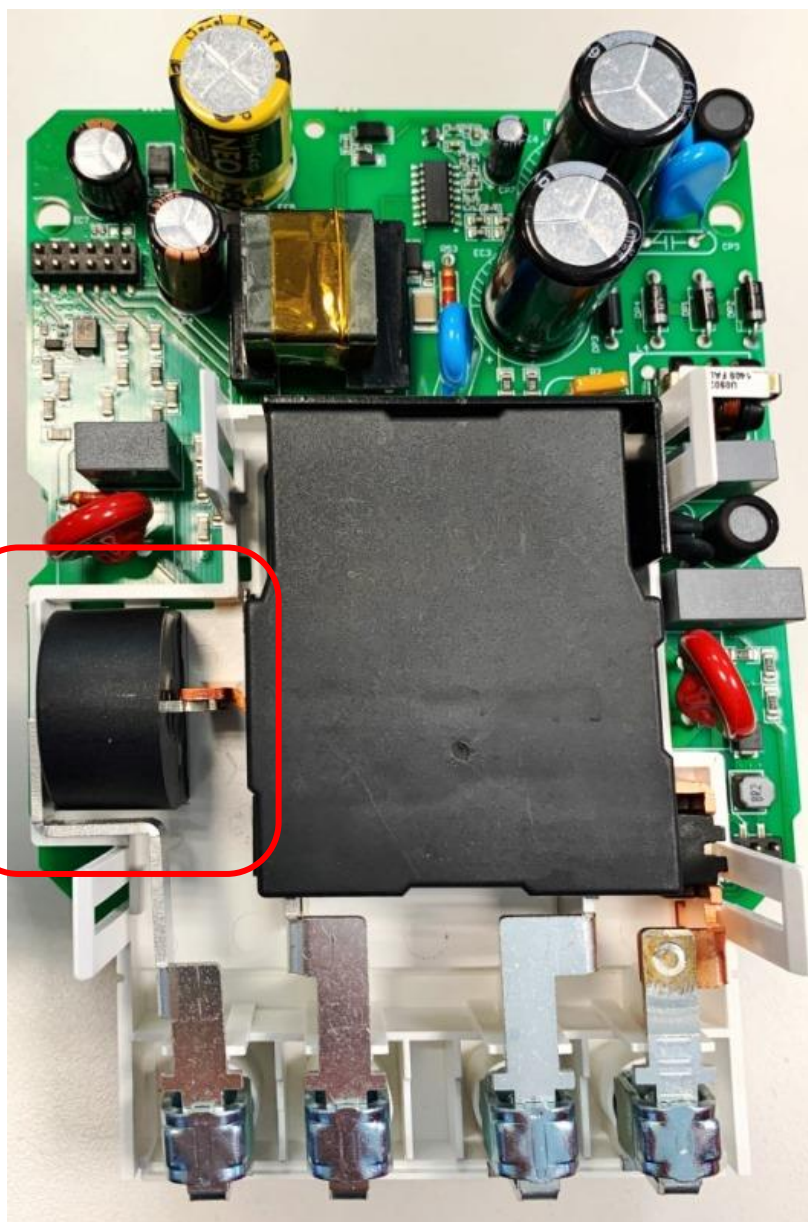
9.3 Current Transformer Description

The Current Transformer is a key component in the design of the power measurement circuits. In the NEXY-M meter, is used to perform current measurement on the neutral line.

The choice of model, material and size of CT determine the characteristics of the current-voltage signal provided to the measurement IC.

Regular checks on the CT is carried out as part of quality control with mechanical strength and electrical characteristics.

The connection with the PCB is obtained without cable (the component is directly welded on the PCB, as it is possible to see in the next pictures.



Considering that the I_{max} of the NEXY-M can be 60A or 100A, two different size of current transformer can be used during the assembling process: one up to 60A and one for 100A (note: 100A version will be implemented as a second step).

Four models of CT provided by two different manufacturers can be assembled into the NEXY-M:

- KDT (one model for $I_{max} = 60A$ and another one for $I_{max} = 100A$)
- FALCO (one model for $I_{max} = 60A$ and another one for $I_{max} = 100A$).

9.4 Calibration of the meter

At the end of the manufacturing process each individual meter needs to be calibrated for setting the best accuracy performances.

To carry out this calibration process a dedicated equipment is used; the calibration is carried out in simultaneously on several meters (eg. 40 meters).

As described in section 6 the meter can be configured in factory to work on:

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- Two value of Grid Frequency: 50 Hz or 60 Hz
- One range or One multi range of Reference Voltage: 230 V or 127 V ... 230 V
- Two Meter Constant: 1.000 imp/kWh (1.000 imp/kvarh) or
4.000 imp/kWh (4.000 imp/kvarh)
- Two Maximum current 60 A or 100 A

For this reason, the calibration of the NEXY-M can vary according to the type of production and specific customization for specific market. The data available on the nameplate of the meter, of course, will correctly indicate the grid frequency, the reference voltage, the meter constants for active and reactive energy and the I_{max} .

Example (Voltage = 230V, I_{max} = 60A)

The active energy measurement is calibrated at nominal voltage in n. 2 load conditions as in the following table:

No	Value of current	Cos ϕ	Cal Item
1	10 A	1	VRMS & IRMS
2	10 A	0.5 Q ₁	

After calibration the meter accuracy is verified at the following load conditions:

No	Check point	Check Item
1	230V 60A Unity	Active
2	230 60A cos180°	Active
3	230 5A Unity	Active
4	230 5A cos180°	Active
5	230 5A 60Deg Lag	Active
6	230 5A cosfi= -0,5ind	Active
7	230 5A cosfi=0,8cap	Active
8	230 5A cosfi= -0,8cap	Active
9	230V 0.5A Unity	Active
10	230V 0.5A cos180°	Active
11	230V 0.25A Unity	Active
12	230V 0.25A cos180°	Active

After calibration the meter accuracy is verified at the following load conditions:

No	Check point	Check Item
1	230V 60A PF Zero	Reactive
2	230V 5A PF Zero	Reactive
3	230V 5A sen270°	Reactive
4	230V 5A sen30°	Reactive
5	230V 5A sen150°	Reactive
6	230V 5A sen210°	Reactive
7	230V 5A sen330°	Reactive
8	230V 0.25A PF Zero	Reactive

The accuracy data verified after the calibration are stored in the production database.

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10 Characteristics of Cut-off device

The single-phase meter NEXY-M integrates a cut-off device used to control the power supply. The cut-off device is implemented by a latching relay. The meter is designed in order to integrate latching relays provided by different vendors.

The relay will be provided already assembled with input/output terminals, shunt and Current Transformer.

The re-connection of cut-off device, can be performed in two ways:

- **Locally:** pushing the meter's front button;
- **Remotely:** via remote protocol command or automatically, after the circuit ZLOAD has checked the variation of impedance between phase- neutral or phase – phase conductors (It is considered as a option for the production of meters). It can be performed by the customer opening and closing the main breaker installed after the meter in the customer network.

10.1 Latching relay characteristics

The two poles latching relay with shunt complies with e-distribuzione specification.

Here are the general characteristics of the latching relay:

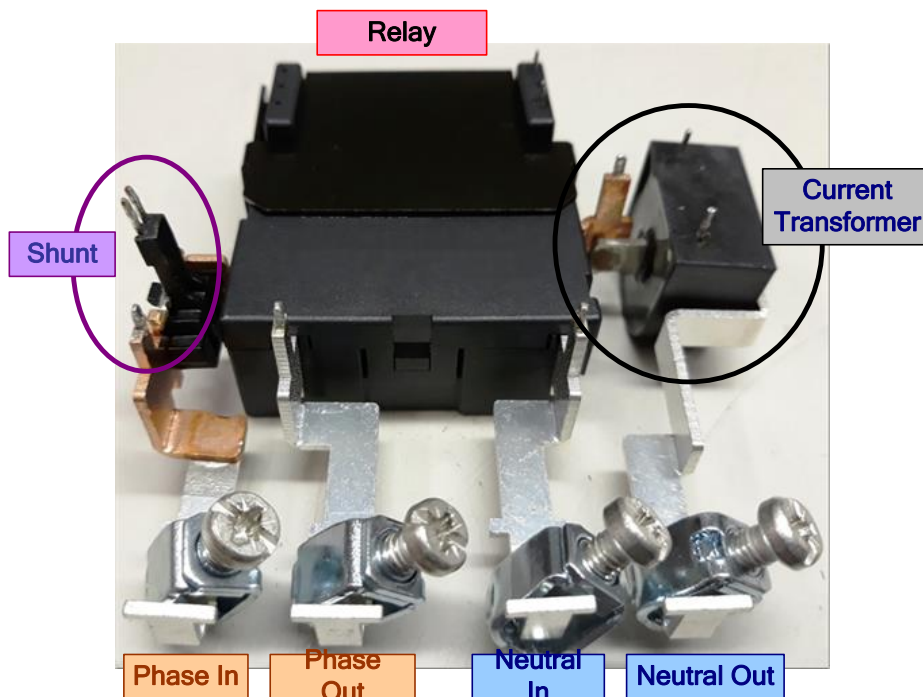
- n. 2 contacts switch (phase and neutral)
- rated current: 80A (for $I_{max} = 60A$) or 100A (for $I_{max} = 100A$)
- rated voltage: up to 230V, ac
- rated frequency: 50Hz or 60 Hz
- resistance across closed contacts: $\leq 0,3 \text{ m}\Omega$
- insulation across open contacts: 2.5 kVrms (1 minute)
- insulation across coil and all contacts: 4 kVrms (1 minute)

environmental conditions:

- temperature range: $-40^{\circ}\text{C} \div 85^{\circ}\text{C}$
- relative humidity: from 5% to 85%

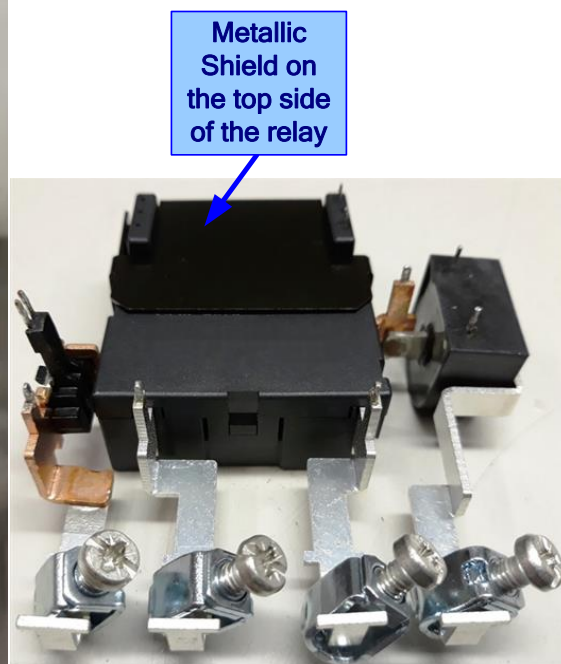
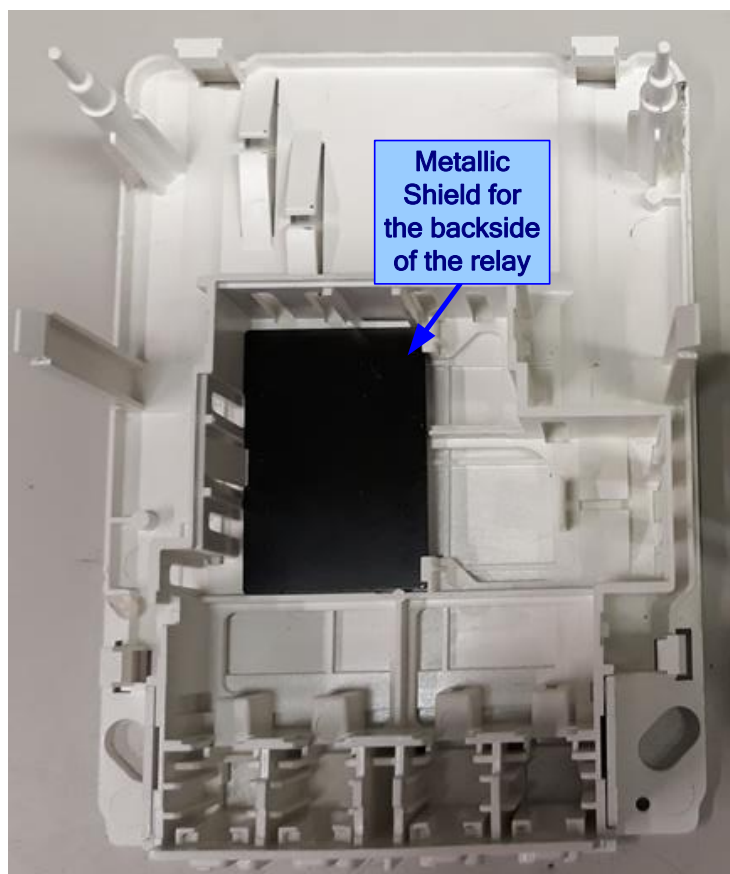
and the general characteristics of the coil driving command:

- rated voltage: 9 Vdc
- driving command duration: $\leq 100 \text{ ms}$



Example of Hong-Fa Latching relay assembled in NEXY-M

The single-phase meter NEXY-M integrates two metallic shields applied on the top side and on the backside of the latching relay used in order to shield the device for the immunity to 1.2T continuous magnetic fields of external origins.

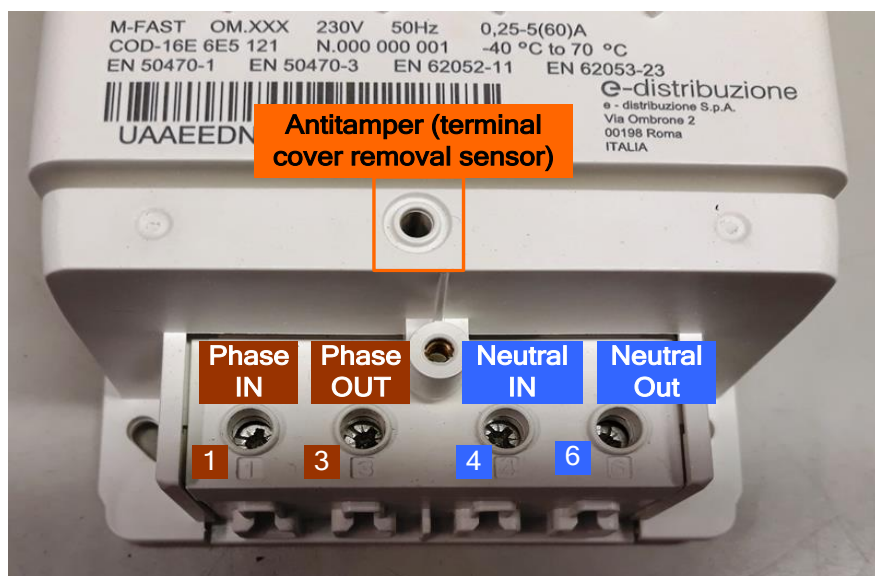
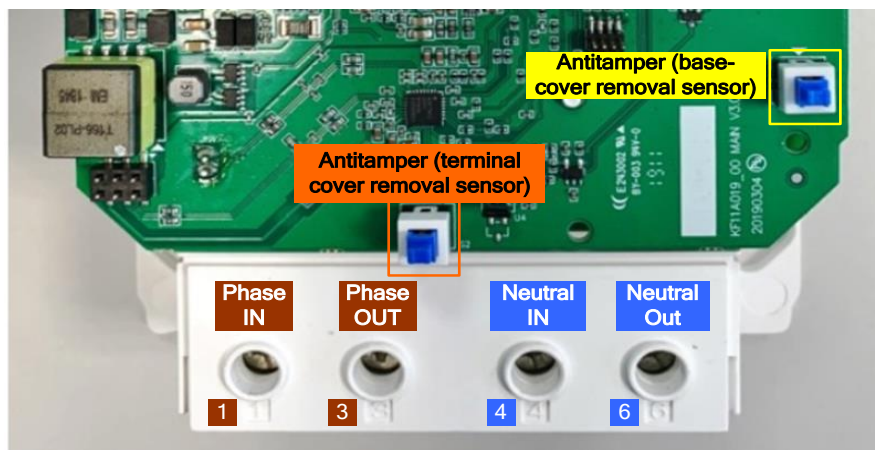


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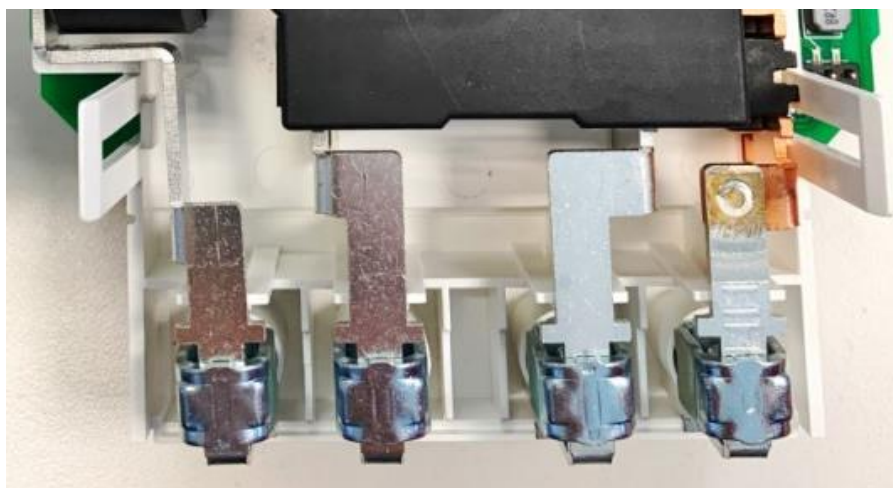
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11 Characteristics of the terminal block

The meter is equipped by a terminal block composed by I/O connections usually adopted in the electricity meters based on DIN 43857 layout. Each connection is equipped by n. 1 screw for tightening the wire by means of a clamp.



Frontal view of terminal block of NEXY-M



Open top view of terminal block of NEXY-M

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About the insulating materials of the terminal block and other relevant parts in close contact with the electrical circuits (e.g. screws support) the following specification are defined (according to EN 50470-1, Section 5.4):

Plastic material: **PC, 10% glass fibre**

e-distribuzione has approved the following products for the manufacturing of the terminal block and case of the NEXY-M meter:

- **STYRON - EMERGE*PC 8701HH**
- **BAYER - Makrolon_9417**
- **SAMSUNG - HN-3104**
- **SABIC - LEXAN_503RS**
- **MITSHUBISHI - Xantar G2F 23 R**

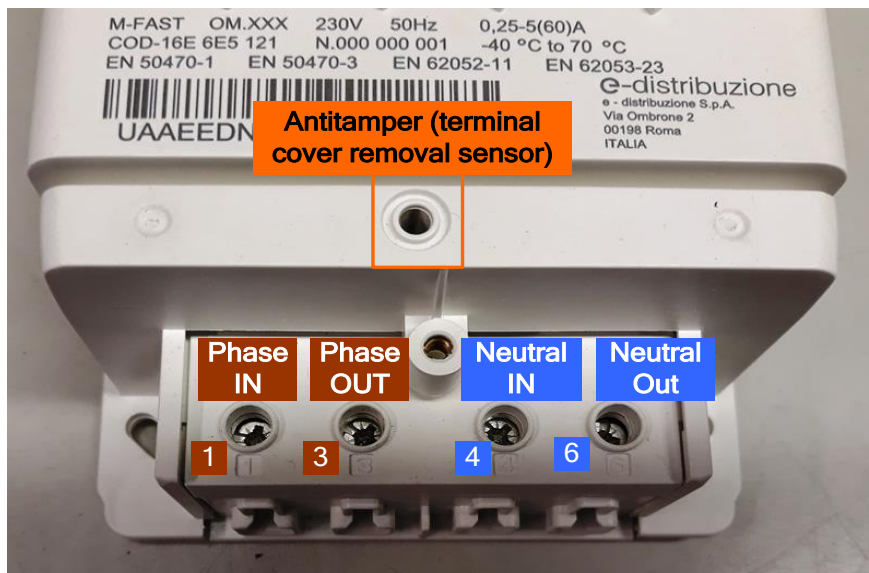
The NEXY-M is able to detect the output voltage even if the cut off device is in open position. This information can be used to activate an alarm to notify a possible failure or fraud condition.

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12 Connection diagram of NEXY-M

The NEXY-M meter has been designed to allow direct connection of LV network line and neutral. It is done by means of terminals located on the front-bottom side of the meter (according to DIN layout).

The connection diagram and the identification of INPUT and OUTPUT of the meter is shown below:

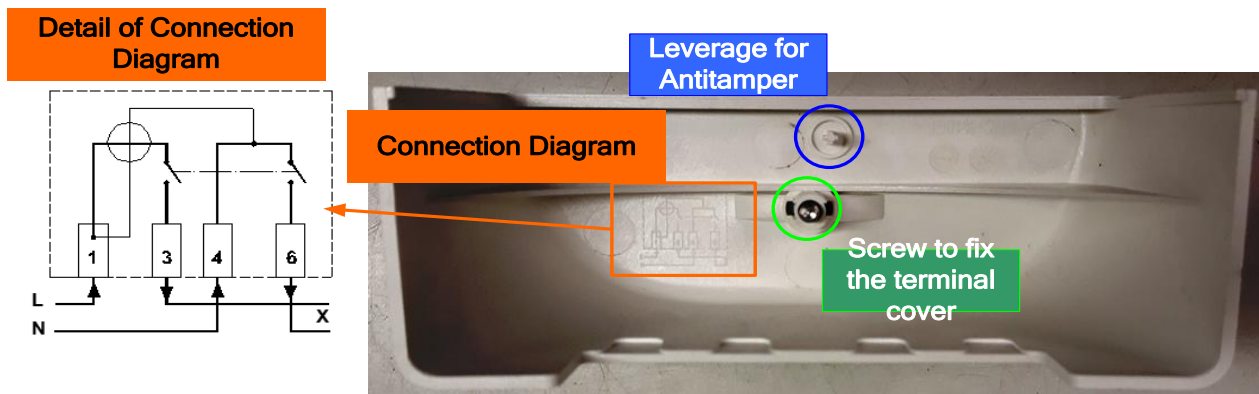


Connection diagram of NEXY-M

Each connection is identified by a number:

- 1 -> Phase line input
- 3 -> Phase line output
- 4 -> Neutral input
- 6 -> Neutral output

The connection diagram is also available of the back side of the terminal cover:

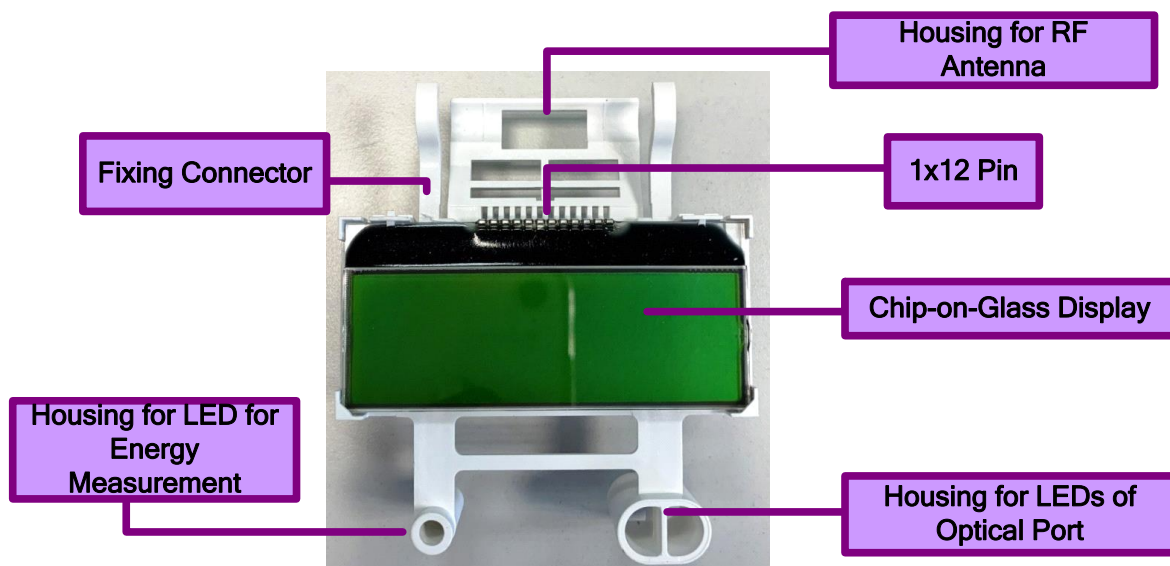


The power supply of the meter is connected before the Measurement IC so that the meter does not measure self-consumption.

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13 LCD module and pulse emitter

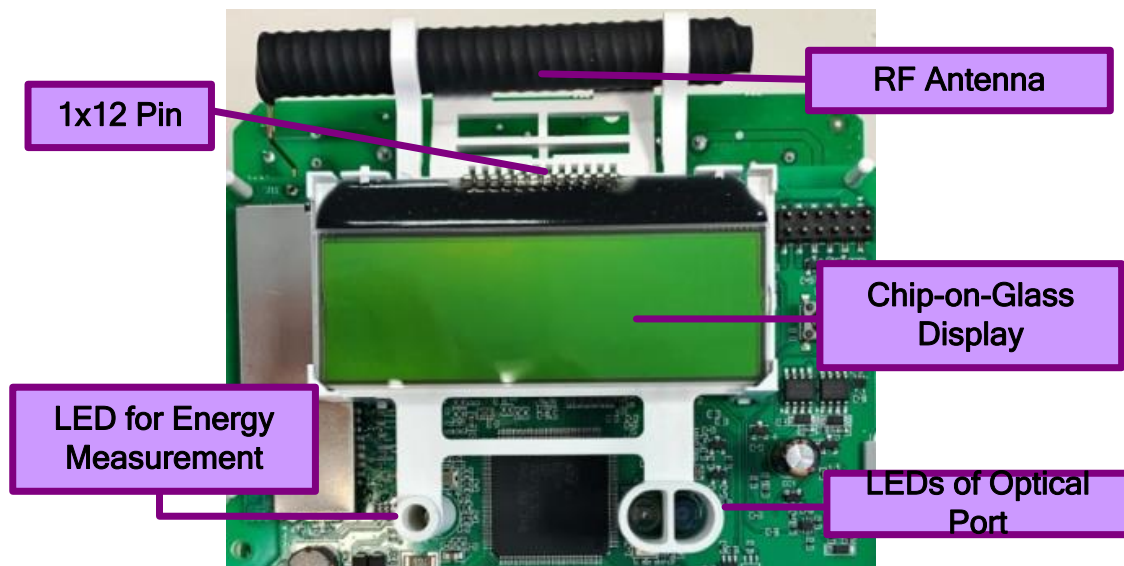
This LCD module is connected to the main PCB with a special fixing base for the Chip-On-Glass display that, thanks to a 12 pin, allow to assemble the LCD on the main PCB with a simple operation that no needs soldering process.



Display installed into the fixing base

The Display fixing base is also equipped with a housing for installing the RF antenna of the radio module (if present) and a protection for LEDS (both pulse emitter and for the optical port).

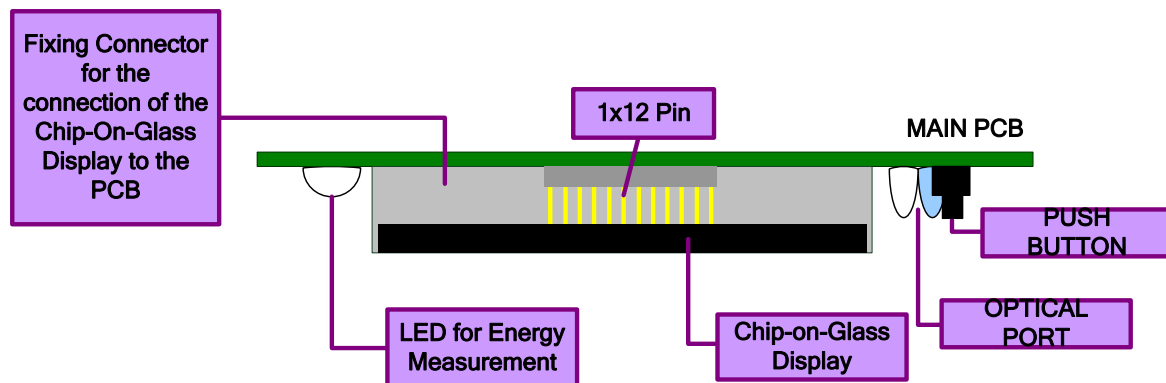
Radio module and LEDS are assembled on the main PCB.



Details of Fixing Base with RF antenna, LEDs and Display installed

The pulse emitter for active and reactive energy measurement is a LED with wavelength and light intensity characteristics those fulfil the requirements described into EN 50470-1, Section 5.11.3.

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The LCD glass has been designed in order to maintain its characteristics (e.g. transparency, strength, etc) in a wide temperature range. It is produced by a quality glass manufacturer and assembled with the electronic module by the same manufacturer or a manufacturer dedicated to that class of process.

The LCD has been designed and produced in order to assure its full functionalities all over the life-time of the meter (15 years).

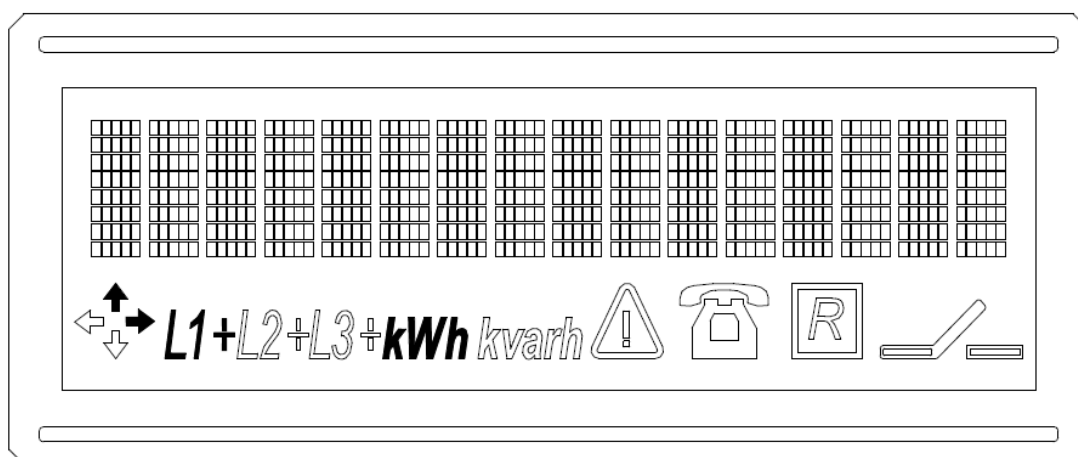
The LCD is not provided by a backlight.

The LCD module is composed of 1x16 alphanumeric characters and 15 icons (special symbols).

Each alphanumeric characters can integrate also a “.”.

For example, the string “1.0.1.8.255” will take 7 characters, because the “.” are included into the characters that precede them.

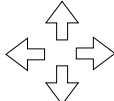
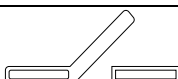
The LCD layout is shown below:



The meaning of icons is described below:

SYMBOL	DESCRIPTION
<i>L1</i>	The phase indicator icon L1 is shown by the meter when the voltage (V _{mis}) is inside of [30% ÷ 100%] of the nominal voltage (127/230V). The icon L1 can not be programmed externally
<i>L2</i>	Not used

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	Not used
	The "+" symbol is shown when the meter measures positive active energy on the line phase. The "-" symbol is shown when the meter measures negative active energy on the line phase.
	This symbol is used to show the energy quadrant in use. Active and Reactive energy arrows are shown only if the energy measurements are higher than the creep values.
	The symbol is shown in the meter when the active power register has been displayed.
	The symbol is shown in the meter together with icon kW when the active energy register has been displayed
	The symbol is shown in the meter when the reactive power register has been displayed.
	The symbol is shown in the meter together with icon kvar when the reactive energy register has been displayed.
	The symbol is shown when active energy pulse emitter is showing reactive energy.
	The warning icon is displayed if any of the programmable flags are set. The icon cannot be programmed externally
	The symbol is shown when the cut-off device results open.
	The symbol is shown according to DLC interface activity and status of commissioning of the meter with the AMM system.

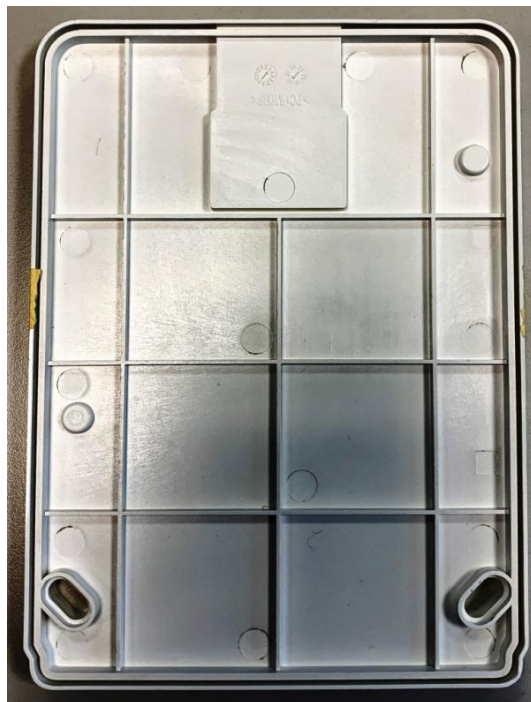
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14 Electromechanical components

The meter is assembled using five principal electromechanical components:

- Two external (external base and external cover);
- Three internal (Internal base, upper and lower supports of relay)

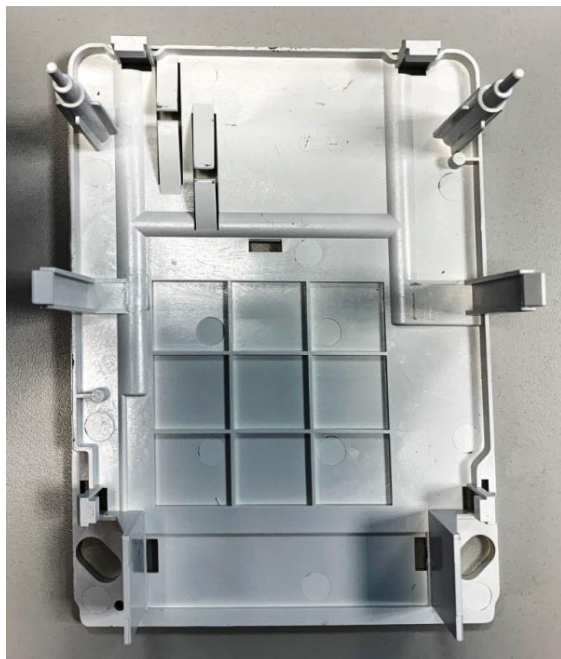
All electromechanical components of the meter are manufactured by using a plastic material PC10%GF.



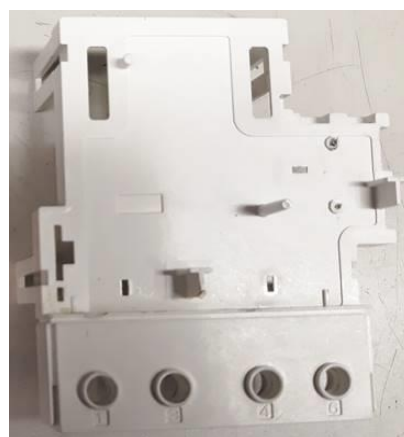
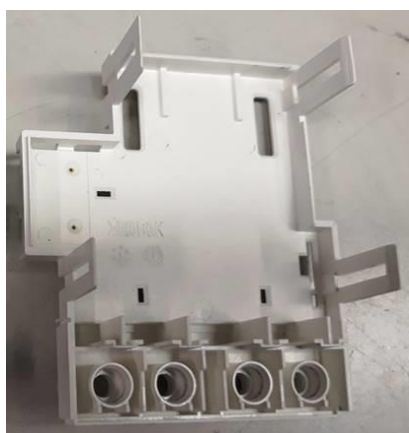
External base: front and rear view



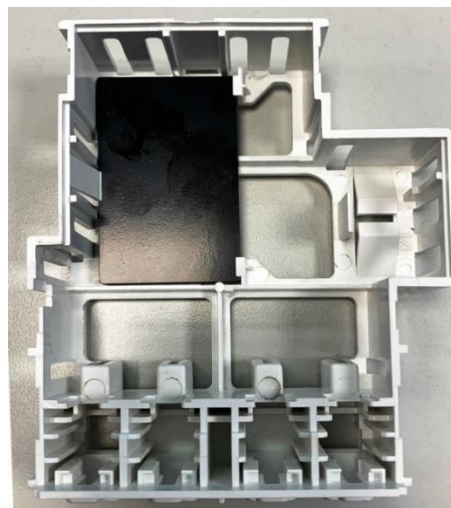
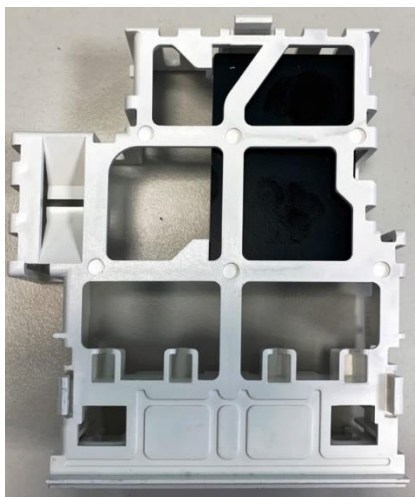
External cover: front and rear view



Internal base: front and rear view



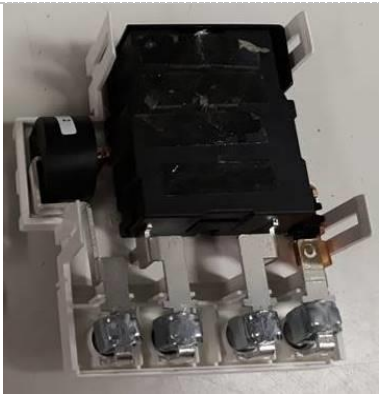
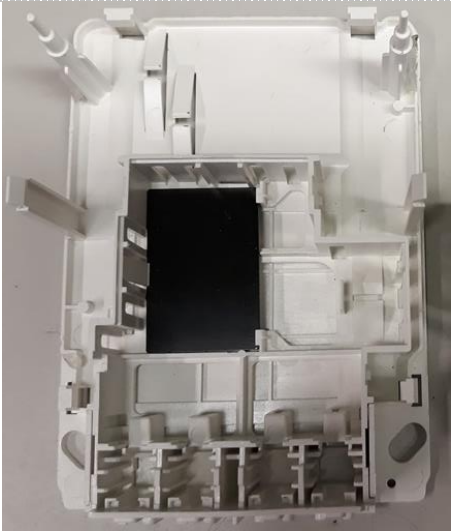
Upper support of relay: front and rear view



Lower Support of Relay: front and rear view

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During the manufacturing process, the meter is assembled in the following way:

1. The relay block is assembled into “upper support of relay” plastic frames without screws but only with plastic latches;	
2. The “the lower support of relay” is assembled into the internal base;	
3. The upper support of relay is assembled with the lower support of relay and internal base	

4. Power Supply PCB and Main PCB are assembled on the plastic frames already formed.



5. Display block is added to the previous step;



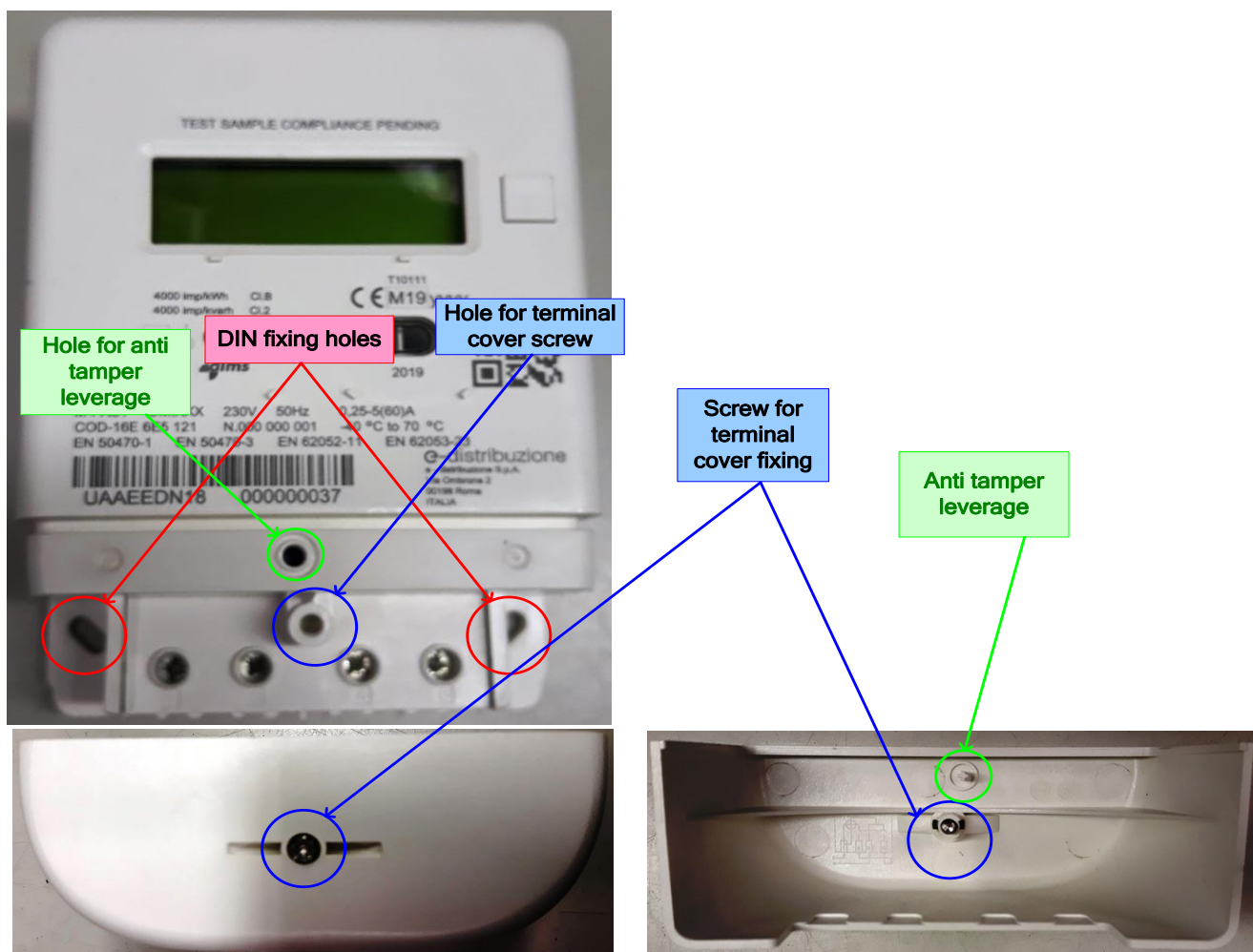
6. The previous block is installed into the external base and then with external cover. Internal and external cover are sealed together using a hot blade welding process.



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After the sealing process the meter cannot be opened without producing damage on the base or cover.

The meter is installed on a DIN socket and fixed with normal screws. These screws are covered by the terminal cover once installed.



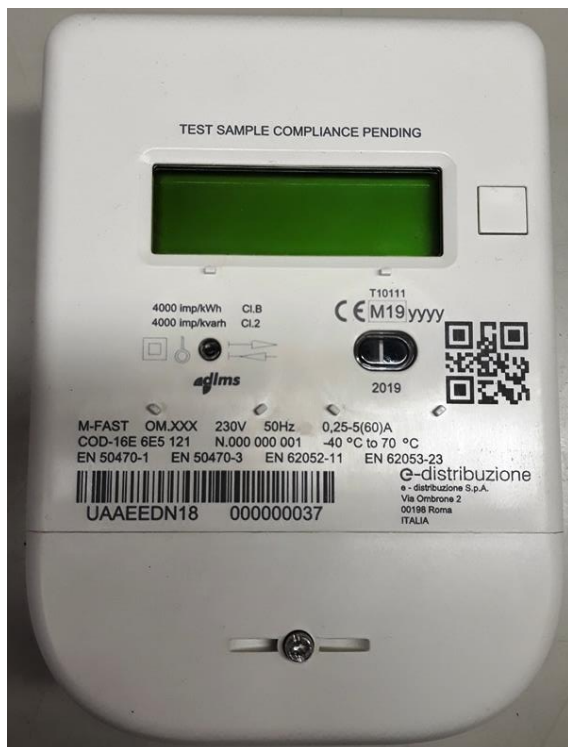
Once installed the terminal cover, an Anti-tamper device is activated in order to prevent abusive handling or meter removing.

The Anti-tamper device produces the notification of an alarm to the central system.

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14.1 Terminal cover

NEXY-M meter can be equipped with the two different terminals cover of the same materials: one “normal” and another one “short”. The short terminal cover useful when the installation location has not much available space. A picture the meter with normal and short terminal cover is shown below.



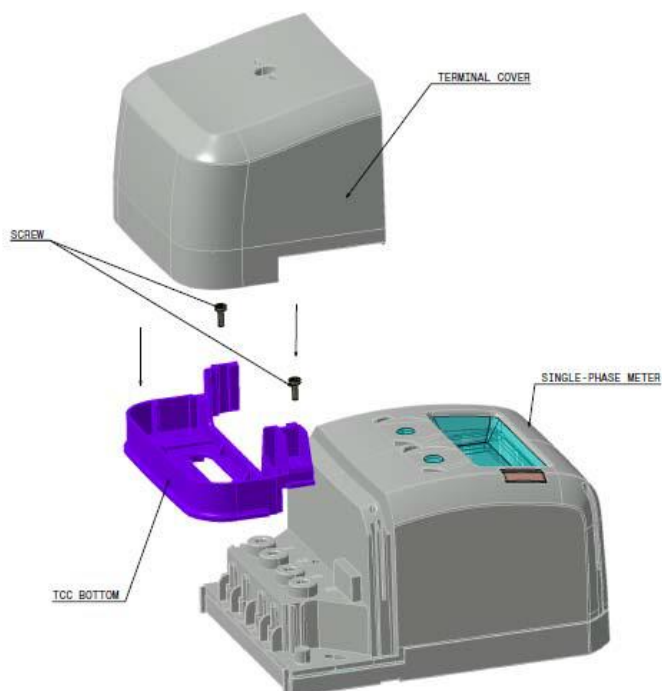
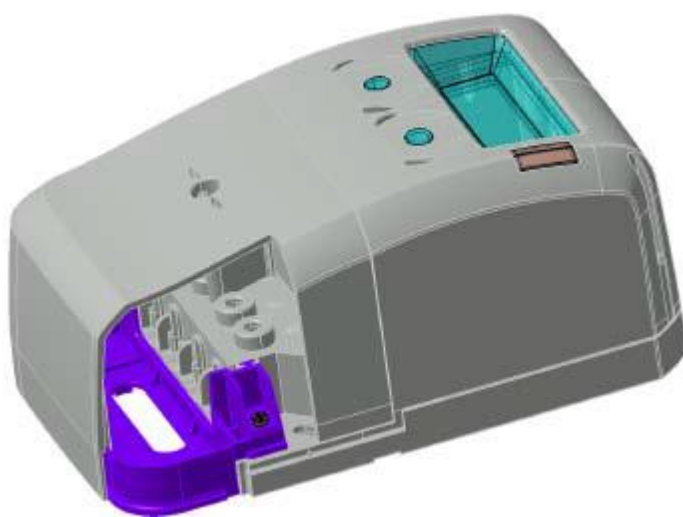
NEXY-M with short terminal cover



NEXY-M with normal terminal cover

14.2 Particular terminal cover (optional)

The NEXY-M can be supplied with this specific back terminal cover to improve the fraud prevention. In the following figure the additional optional part (violet):

**Single parts of the back terminal cover option****General overview**

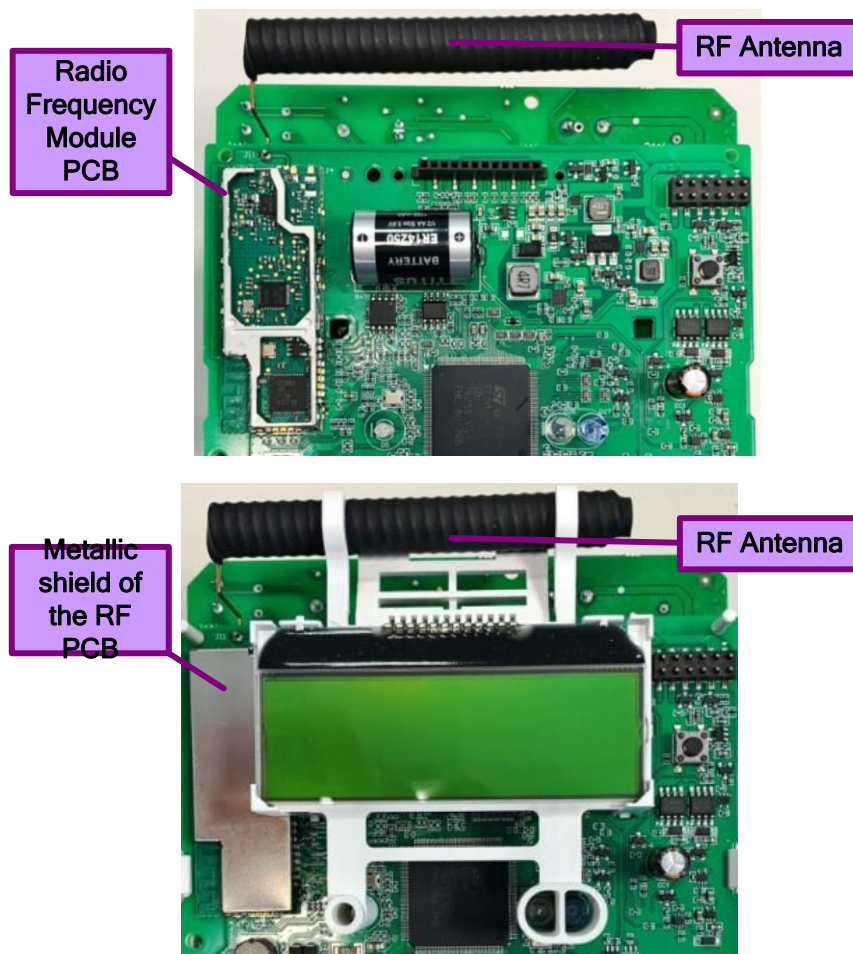
14.3 Radio Modem & Antenna

On the main PCB of NEXY-M, can be installed an optional Radio Module that integrates Radio Frequency @ 169 MHz WMBUS and Bluetooth communication capabilities. This module can be used to send spontaneous message @ 169 MHz or to be used as remote (@169 MHz) or local (via Bluetooth) back-up channel respect to the PLC or ZVEI.

This module is assembled during the manufacturing process into e-distribuzione's factory.

The Radio Module, if installed, is connected to the RF antenna and it is covered by means a protective shield.

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All the information about the functionality connected to the radio communication are detailed in a separate document, related to the SW functionalities of the meter.

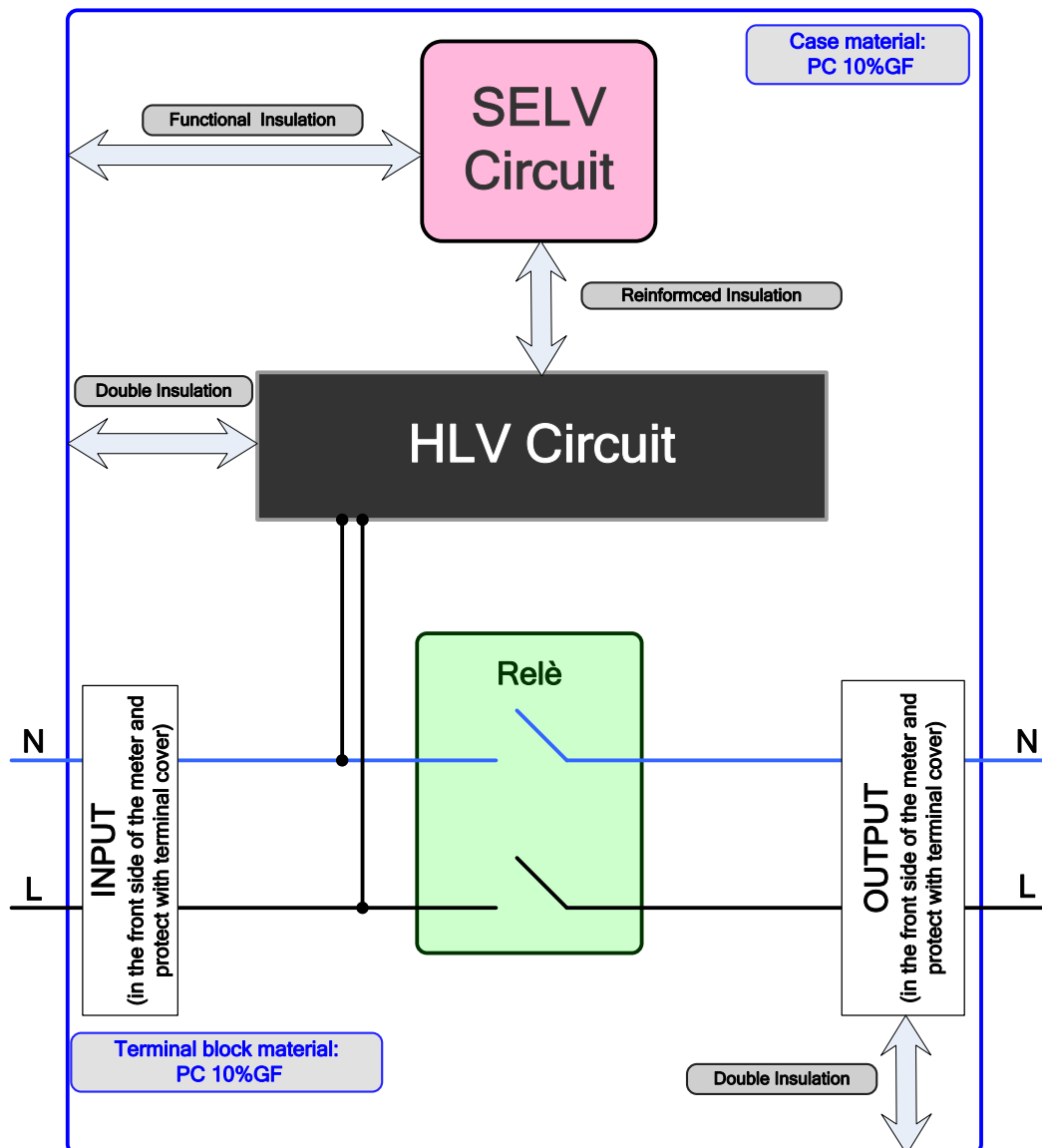
14.4 Insulation diagram between meter's parts

NEXY-M has been designed in order to have no accessible parts under voltage (neither HLV nor ELV/SELV).

Furthermore, input and output terminals are located on the front side of the meter and they are designed to be considered as “no accessible” part (according to definition of IEC 62052-31).

During normal use of the meter, terminals are protected with a terminal cover that cannot be removed without activating an alarm.

The following figure shows the insulation diagram between NEXY-M parts:



15 Manufacturing process overview

In this paragraph an overview of the manufacturing process will be described:

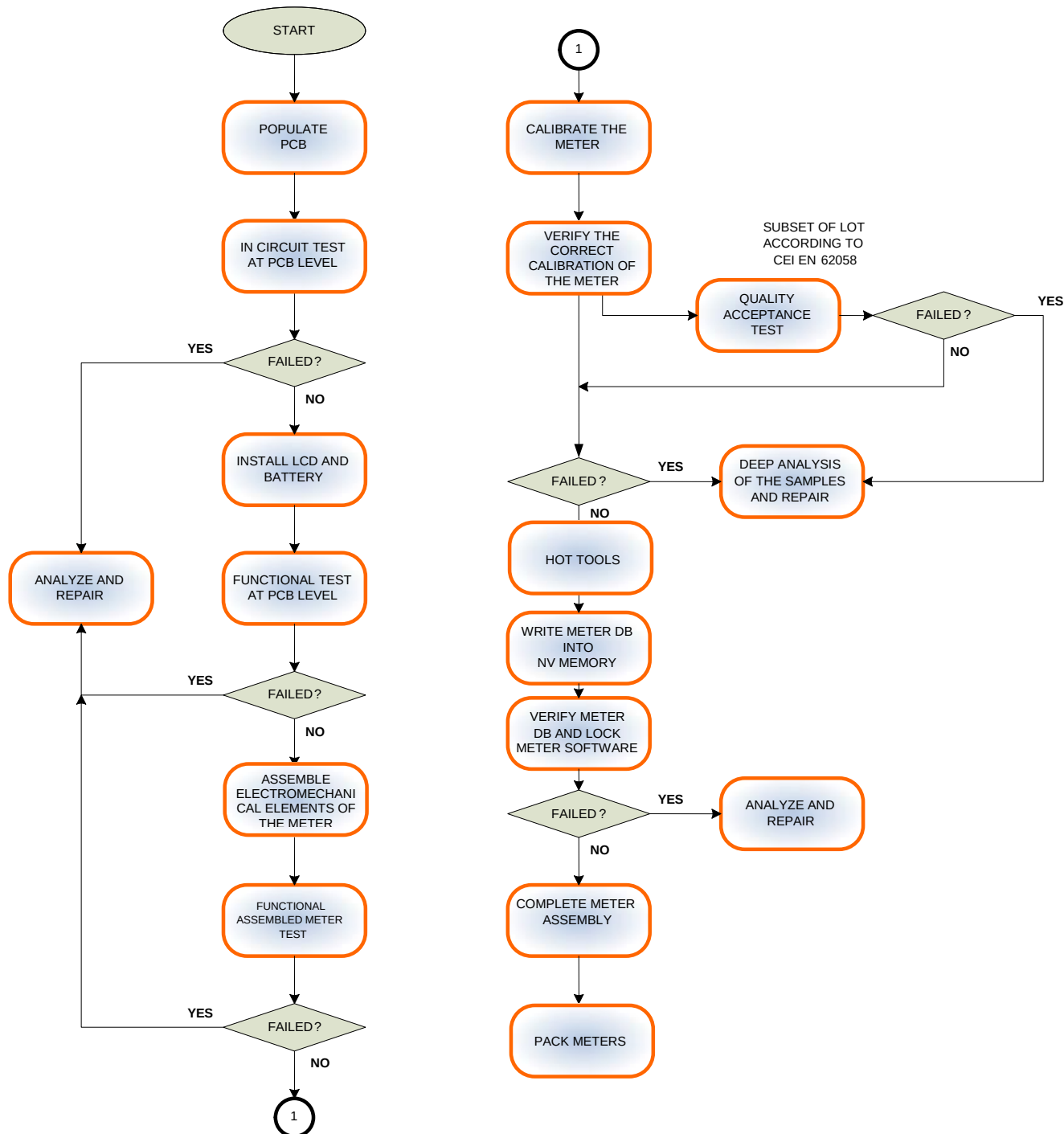


Fig. 2 – Manufacturing process flow

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15.1 Populate PCB

The two (main and Power supply) PCBs are populated by almost all electronic components (after a bare PCB test in order to detect manufacturing defects prior to population).

At this stage, the key components of the meter are selected for storage of their batch numbers into the production DB. This DB provides the manufacturer with an identity card of each single meter.

The list of components selected for the traceability is described in next paragraph.

15.2 In Circuit Test at PCB level

The assembled PCB is tested by means of a built-for-purpose machine in order to verify the tolerance and the correct insertion of the candidate components.

If the test fails, the PCB is analysed and repaired.

At the same stage, the following operations are performed:

- MCU and FLASH programming
- EEPROM programming
- RTC calibration
- Verify local FLAG communication

The result of each operation is stored into the production DB.

15.3 Functional Card Test at PCB level

The assembled PCB, now equipped also by LCD and battery, is tested by means of a built-for-purpose machine in order to verify the correct operation of all elements.

The following testing are performed:

- Power supply
- Optical interface
- Measurement IC
- RTC
- PLC
- Tripping command
- Tamper circuit
- External Flash
- LCD module
- Push Button
- LED
- RF

If the test fails, the PCBs are analysed and repaired.

The result of each testing is stored into the production DB.

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15.4 Functional Assembled Meter Test

The assembled meter, now equipped also by all electromechanical components, is tested by means of a built-for-purpose machine in order to verify the correct operation of all elements.

The following testing are performed:

- FLAG communication
- DLC communication
- RF communication (@169 MHz & Bluetooth)
- PUSH button
- Tamper switches
- Meter integrated cut-off device
- SHUNT circuit
- RTC
- Neutral measurement calibration
- Supercap
- RTC

If the test fails, the PCB is analysed and repaired.

The result of each testing is stored into the production DB.

15.5 Calibration of the Meter

The assembled meter is calibrated against certified energy standards by means of a built-for-purpose machine equipped by reference measurement instruments.

This operation is performed following the procedure described into Par. 9.4.

The result of calibration is stored into the production DB.

15.6 Verification of the Meter Calibration

Calibration of each meter is then verified by means of a specific machine in order to assure the required accuracy in energy/power measurement.

If the test fails, the meter is analysed and re-calibrated.

This process has been designed in order to absolutely avoid that a meter without the required calibration performances is accepted and delivered to the shipping.

The result of verification is stored into the production DB.

15.7 Quality Acceptance Test

The Quality Acceptance Test is performed on a subset of all produced meter (also called “lot”) according to CEI EN62058-31

If the test fails, the subset of lot is deeply analysed.

The result of each testing is stored into the production DB.

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15.8 Hot tools

The mechanical seal of the meter is performed by means of hot blade tools. In particular the base and cover are sealed together by means of hot blade welding process.

After the two steps the meter results completely sealed and closed.

15.9 Configuration of Meter DB

After calibration and Quality Acceptance test, the DB of the meter (that includes all functional parameters and that is stored into non-volatile memory) is configured.

The result of configuration is stored into the production DB.

15.10 Verification of Meter DB and software Lock

The DB of the meter is verified in order to assure the correctness of all data written during configuration.

The result of verification is stored into the production DB.

If the verification is ok, the meter software is locked in order to avoid any change from now on.

15.11 Complete Assembly of Meter

At the last stage of the production process, the assembly of the meter is completed by sealing the meter, installing the sealing screws, additional plates and configuring the meter for the first in field installation.

Meters are then packaged for shipping.

16 Meter key components included into traceability

During the production process, all components of the meter are identified for traceability.

In the following paragraphs, meter's components for traceability during manufacturing are described:

16.1 Electronic and Mechanical Components

	Schematic Reference	Description/Identification
Main PCB	/	KF11A019_00 MAIN V3.0 20190304
Power Supply PCB	/	KF11A019_00 POWER V3.0 20190306
Measurement IC	U10	Analog Device “ADE7953”
Shunt	/	Manganine (the composition is CU-86 & Mn-12 & Ni-2). Shunt resistance across measuring contact: - 280 $\mu\Omega$ (if the I_{max} = 60A) or, - 184 $\mu\Omega$ (if the I_{max} = 100A)
Current Transformer	/	60A: FDC-KDT-218 or FALCO T81018 100A: FDC-KDT-216C or FALCO T90069
Power Transformer	VT1	EP-17 switching transformer N1:N2:N3:N4=70:10:13:70 (KD, FALCO)
Microcontroller	U7	32 bit with 1Mbit of embedded Flash memory and Multimodulation PLC Modem “ST ECOM2”

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		Please note that due to security aspects, this component can be marked ST ECOM2 or not. In this case its part number is ED1536-x (without ST logo).
PLC Modem	/	Embedded into MCU
EEPROM	U8	512kbit EEPROM: (ST, Renesas, Microchip, Onsemi, Belling)
Non Volatile Flash Memory	U9	8 Mbit (Winbond, Macronix, Issi, GD, Huahong, Cypress, Adesto, Amic)
RF Module	/	Custom Radio Module 169 MHz BLE Board (ID 15016855-02)
RF Antenna	/	169 MHz stick Antenna PROANT PRO-IT-586
Battery	BAT1	1/2AA Lithium Battery 3,0 Volt $\geq$ 850mAh or 3,6 Volt & $\geq$ 1200mAh (FDK, VARTA, SAFT, TITUS, VITZROCELL, EEMB, ACT)
LCD Module	/	Chip on Glass (VARITRONIX, TRULY, SUCCESS, YEEBO, HUARI, HOLITECH, AV Displays, Rifda, Techshine, Smartwin)
Magnetic sensor	U4	High sensitivity micropower omnipolar Hall-effect switch
Relay	/	Hong-Fa (Please refer to section 10.1)

16.2 Firmware

As described in section 7, the NEXY-M can be equipped with two different type of software: one that supports DLMS communication protocol and one that supports Meters and More communication protocol.

In both cases, the meter is designed to support equipped with a built-for-purpose firmware that is organized in three different portions:

- **fixed part:** this part of the firmware implements the boot procedure
- **legally relevant downloadable part:** this part of the firmware implements the legally relevant functionalities
- **non-legally relevant downloadable part:** this part of the firmware implements the non-legally relevant functionalities

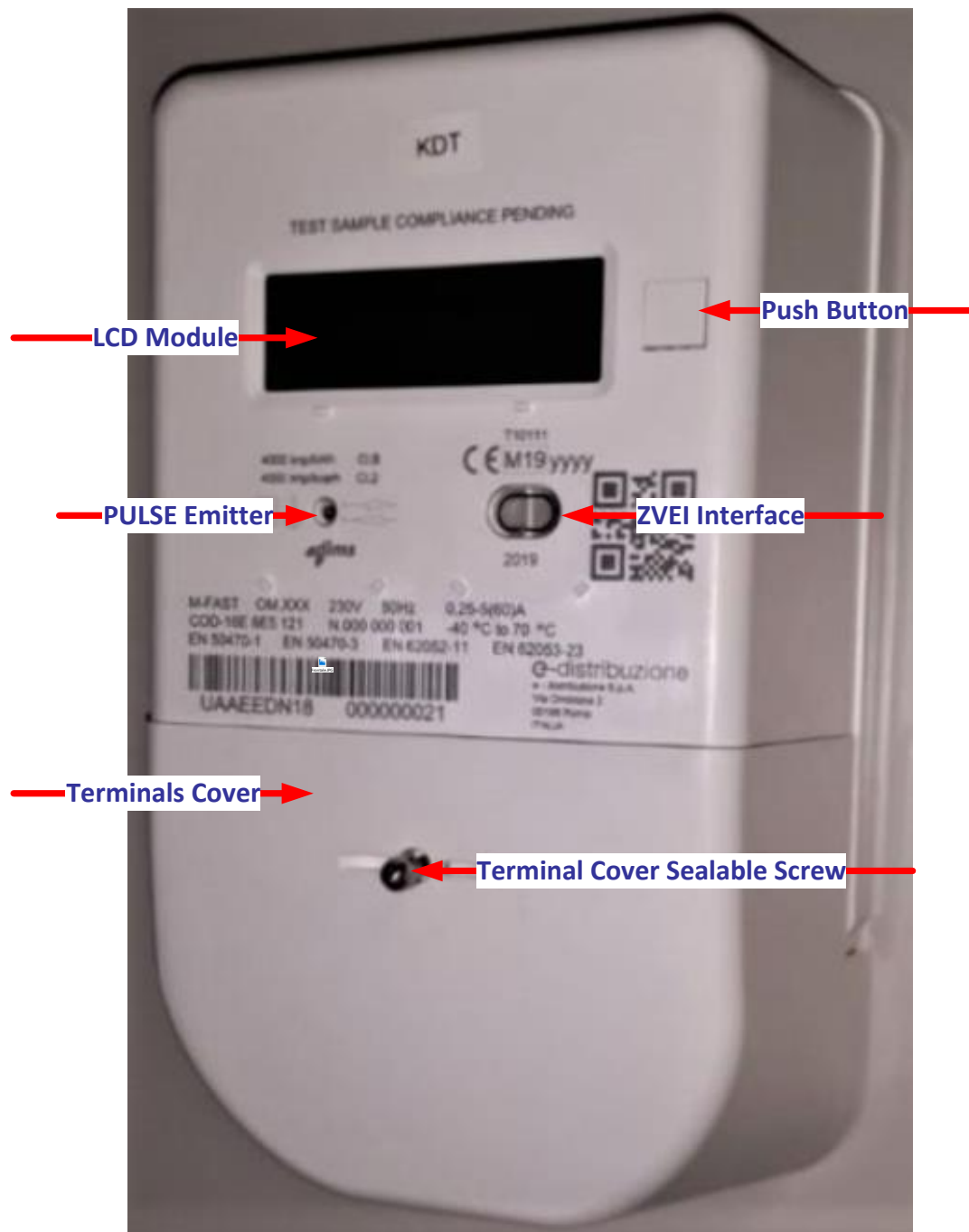
It is designed in order to provide the possibility to update the downloadable parts of the software by means of a specific and secured procedure (an authentication request has to be performed before starting the download of a new code).

Also the firmware is traced during the manufacturing process.

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17 NEXY-M front side description and dimension

The following picture describes the front side of the NEXY-M meter.



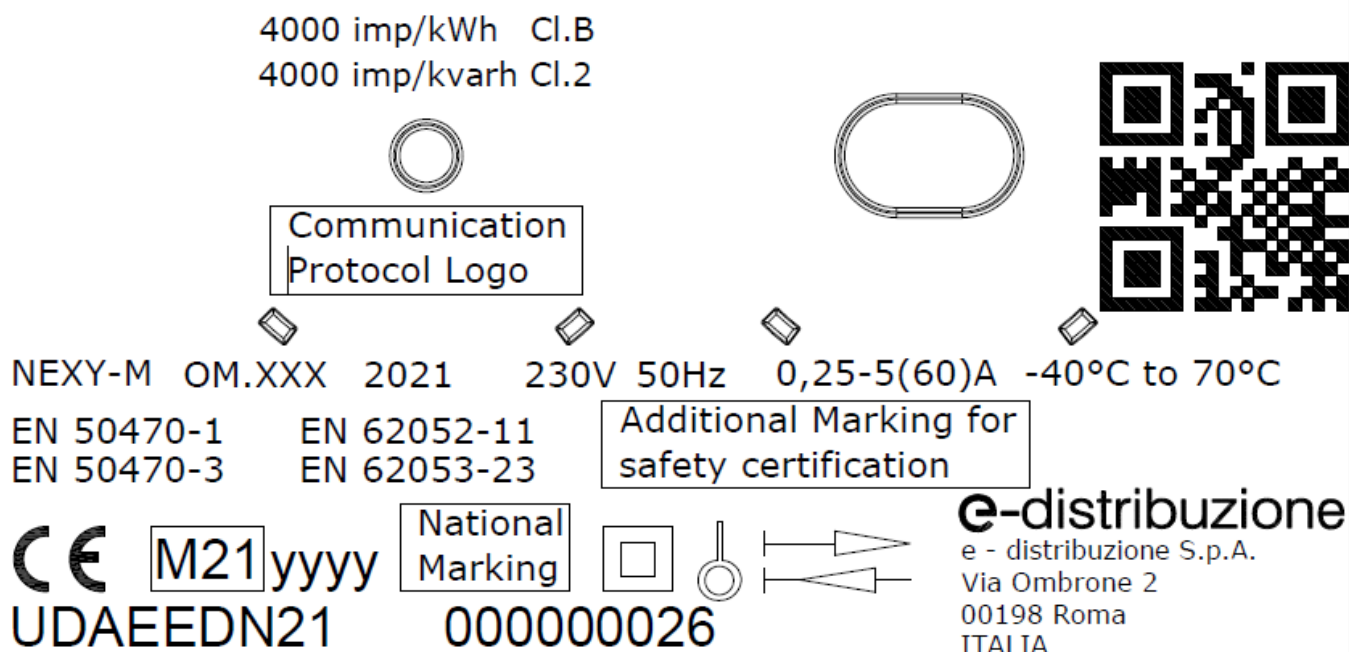
The dimension of the NEXY-M meter are summarized in the following table:

	NORMAL TERMINAL COVER	SHORT TERMINAL COVER
HEIGHT		
WIDTH		
DEPTH		

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18 NEXY-M markings and indicators

The nameplate layout and sealing of NEXY-M complies with e-distribuzione specification. The name plate position on the single phase meter cover is described in the following

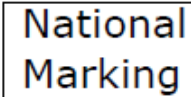
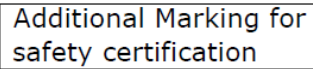
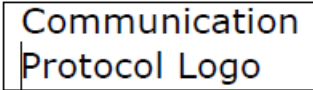


Example of NEXY-M nameplate

The nameplate reported in the previous picture is an example that includes relevant and optional information. Other possible layout including additional information can be possible assuring that relevant information according to MID will be always present.

The explanation of the information included in the previous nameplate is the following:

Nameplate data	Description	Relevant
NEXY-M	Meter type (Single Phase Electronic Meter)	Yes
OM. XXX	e-distribuzione Homologation number	No
2021	Year of manufacture of the meter	Yes
230 V	Reference voltage This field may vary according to the factory configuration	Yes
50 Hz	Reference frequency (Hertz) This field may vary according to the factory configuration	Yes
0,25-5(60) A	Imin, Iref and Imax current of the measuring system This field may vary according to the factory configuration	Yes
UAAEEDN21 000000026	Meter code and serial number	Yes
4000 imp/kWh	Meter constant active energy This field may vary according to the factory configuration	Yes

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4000 imp/kvarh	Meter constant reactive energy This field may vary according to the factory configuration	Yes
Cl. B	Meter class indices active energy	Yes
Cl. 2	Meter class indices reactive energy	Yes
-40° ÷ +70°	Operating Temperature Range	Yes
EN 50470-1	Electricity metering equipment (AC) – General requirements, tests and test condition – Metering equipment (class indexes A, B and C)	No
EN 50470-3	Electricity metering equipment (AC) – Particular requirements – Static meters for active energy (class indexes A, B and C)	No
EN 62052-11	Electricity metering equipment (AC) – General requirements, tests and test condition	No
EN 62053-23	Static meter for measurement of electrical reactive energy for alternate current.	No
	Marking CE	Yes
M21	MID Marking with year of application of MID marking	Yes
YYYY	Code of the Notified Body that has issued the MID Module D Certificate	Yes
	Space reserved for national type approval marking (if any)	Depending on the installation country
	Meter of protective class II	Yes
	Meter graphic symbol with one measuring element	Yes
	Bi-directional capability	Yes
e-distribuzione	Meter Manufacturer Registered trade mark	Yes
e - distribuzione S.p.A. Via Ombrone 2 00198 Roma ITALIA	Meter Manufacturer Name and postal address at which the manufacturer can be contacted.	Yes
	Space reserved to safety marking (To be updated)	Yes
	Space reserved to communication protocol Marking	No

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Connections scheme and terminal marking of the NEXY-M comply with European standard .

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19 PCBs assembly details

For reliability reasons the PCBs have a gold plated finish. This reduces the risk of oxidisation and corrosion over time and also provides better quality soldering during manufacturing.

The drawings that follow are for reference only and do not contain all of the relevant information needed for the assembling of the NEXY-M.

The list of drawings includes:

- Top Layer routing (Main PCB, Power Supply PCB and RF PCB).
- Bottom Layer routing (Main PCB, Power Supply PCB and RF PCB).
- 2 Inner Layer routing (Main PCB)
- Silk Screen with component designators (Main PCB, Power Supply PCB and RF PCB).

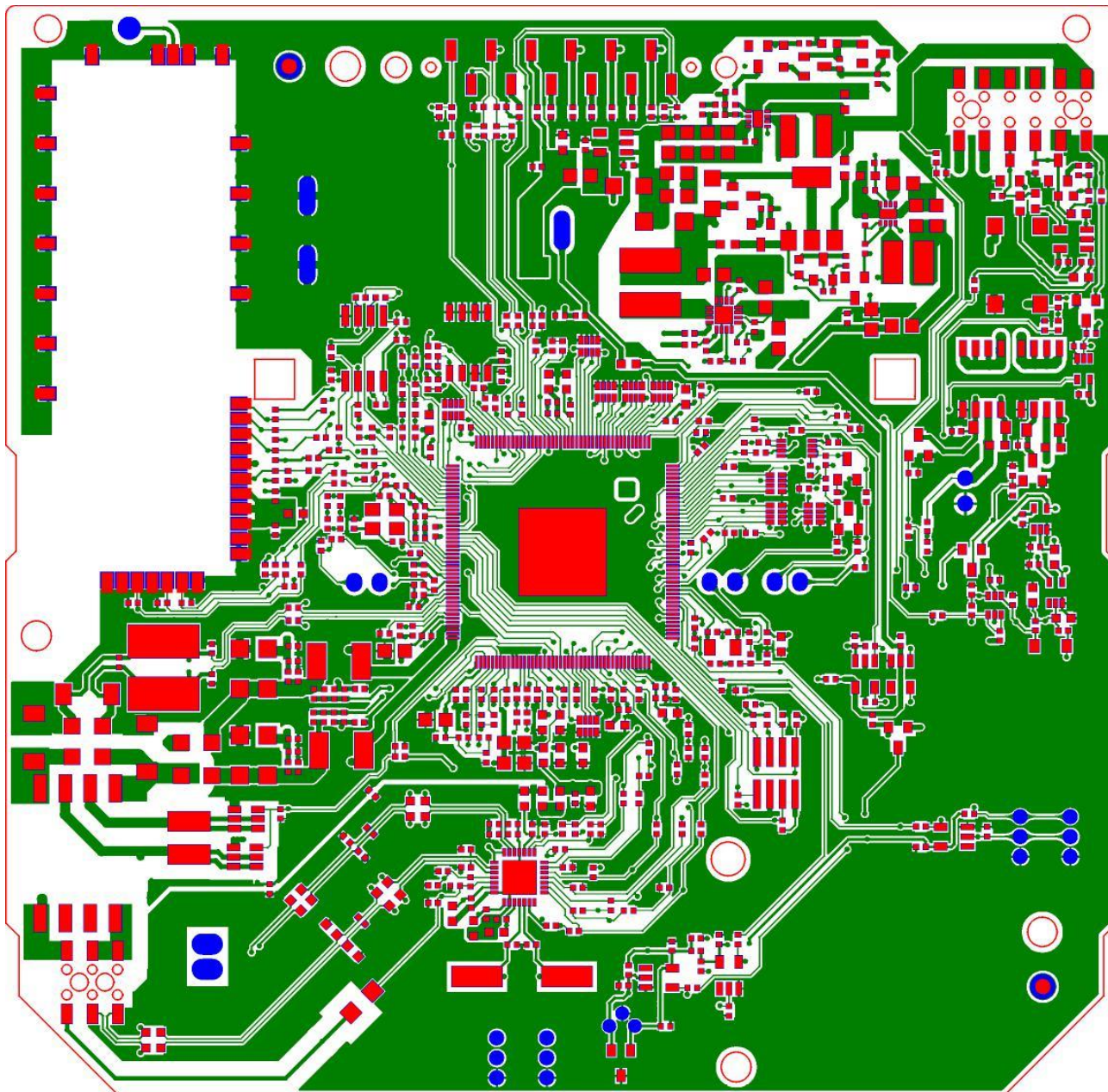
Drawings of the main PCB have been designed in order to allow the fitting of all components of the meter. As reported in the previous paragraphs, some of these components are optional and other are not installed in this phase of the project.

In particular:

- Circuitry related to the connection of the RF Module PCB is to be considered as “optional”, so it can be installed or not;

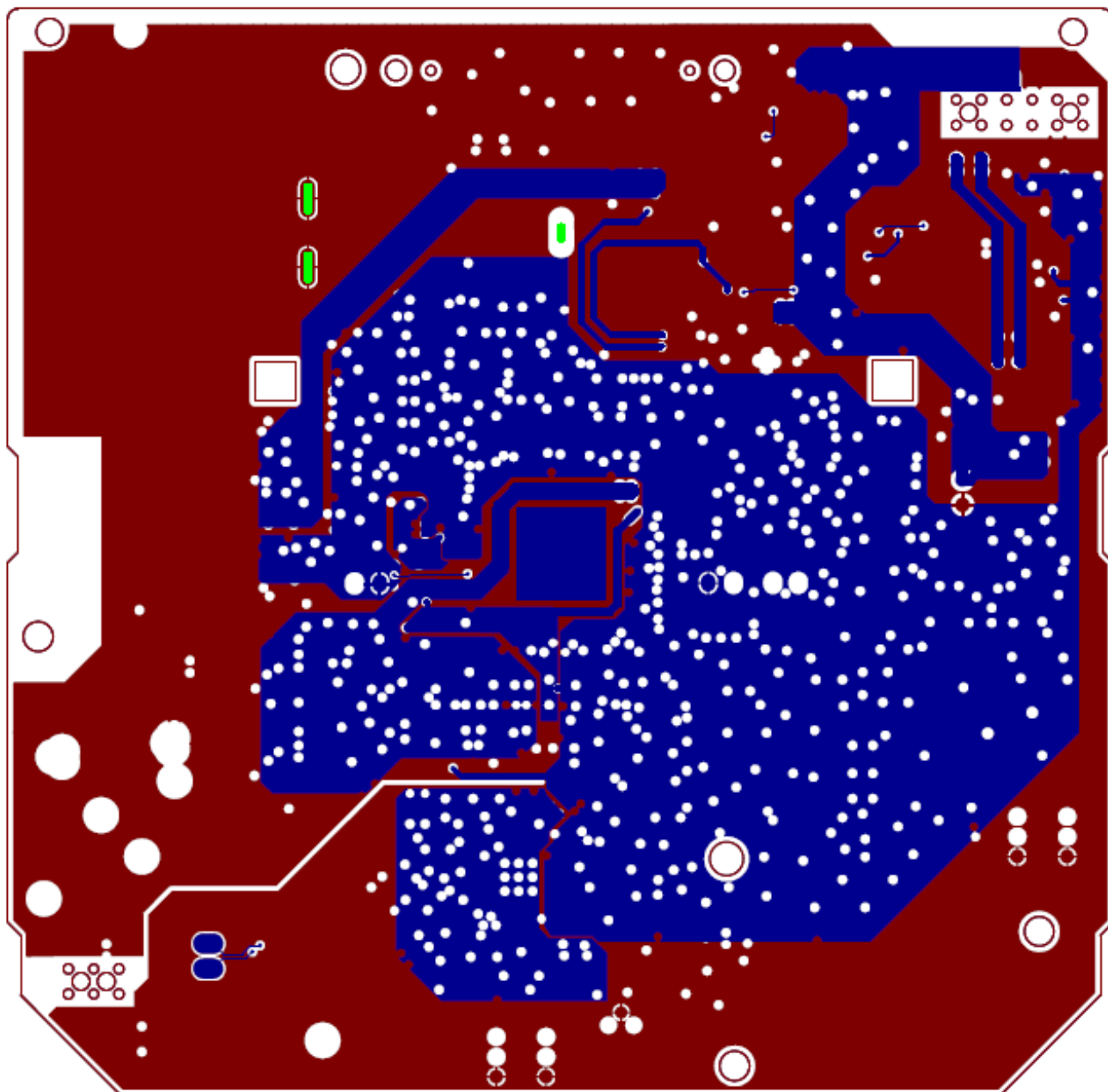
“RF Module PCB” is optional, so the meter can be assembled with or without this PCB and the RF Antenna.

The RF antenna is not designed and assembled by e-distribuzione, but is bought from an external supplier “as is”, so drawing are not available.

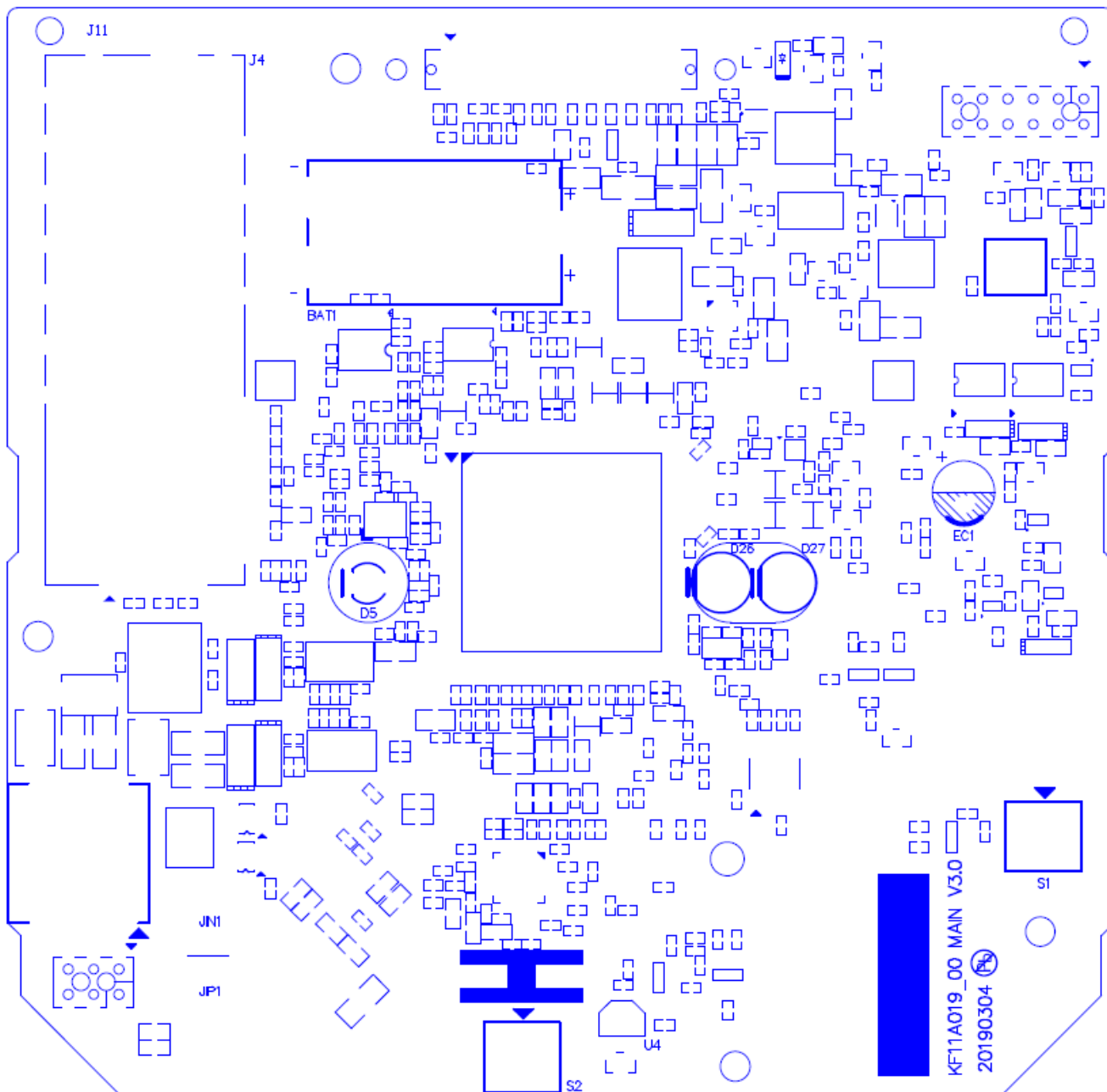
19.1 Main PCB (KF11A019_00 MAN V3.0 20190304)**19.1.1 Top layer routing (KF11A019_00 MAN V3.0 20190304)**

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19.1.3 Inner layer routing (KF11A019_00 MAN V3.0 20190304)

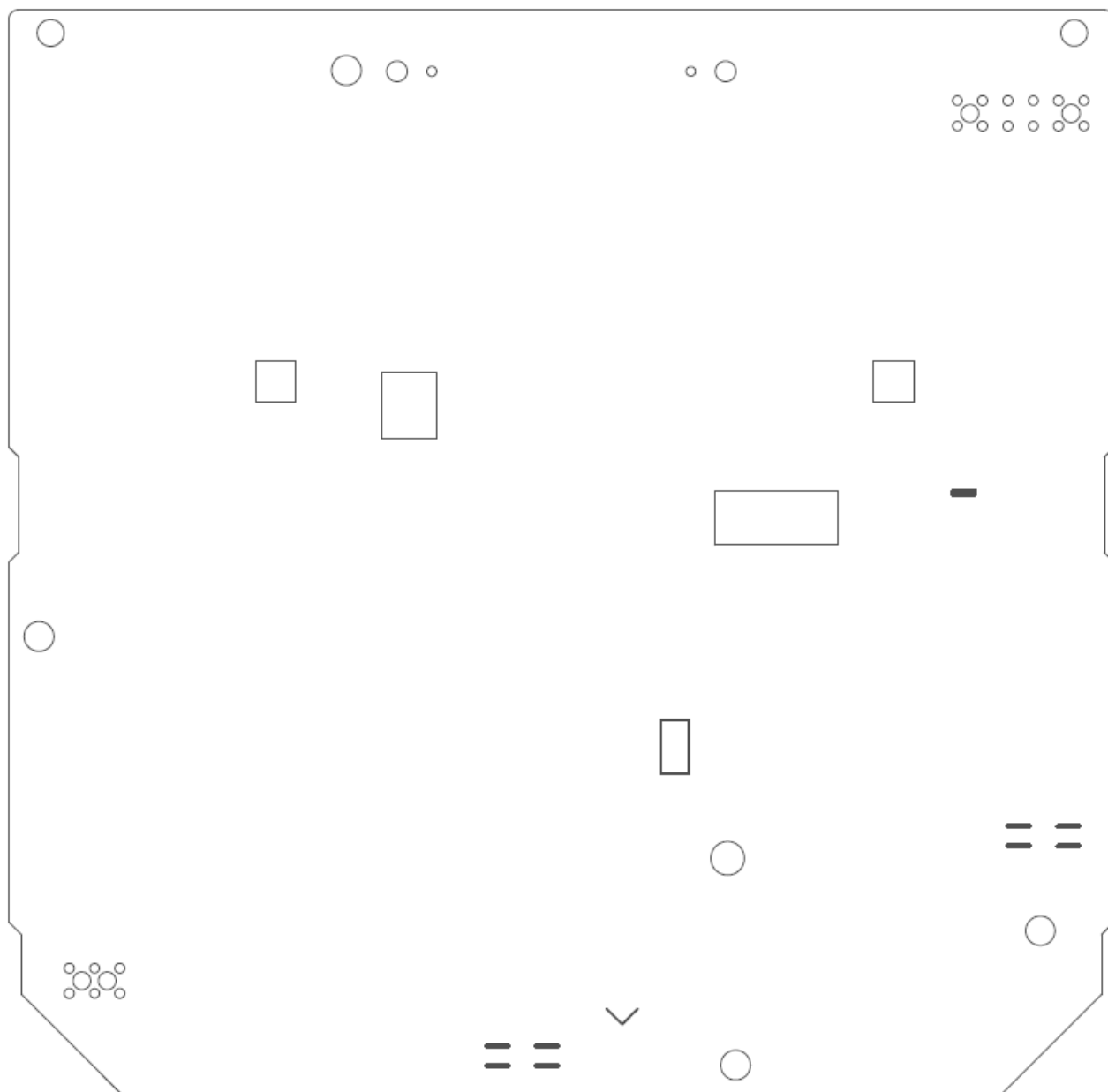


19.1.4 Top Silk Screen with component (KF11A019_00 MAN V3.0 20190304)



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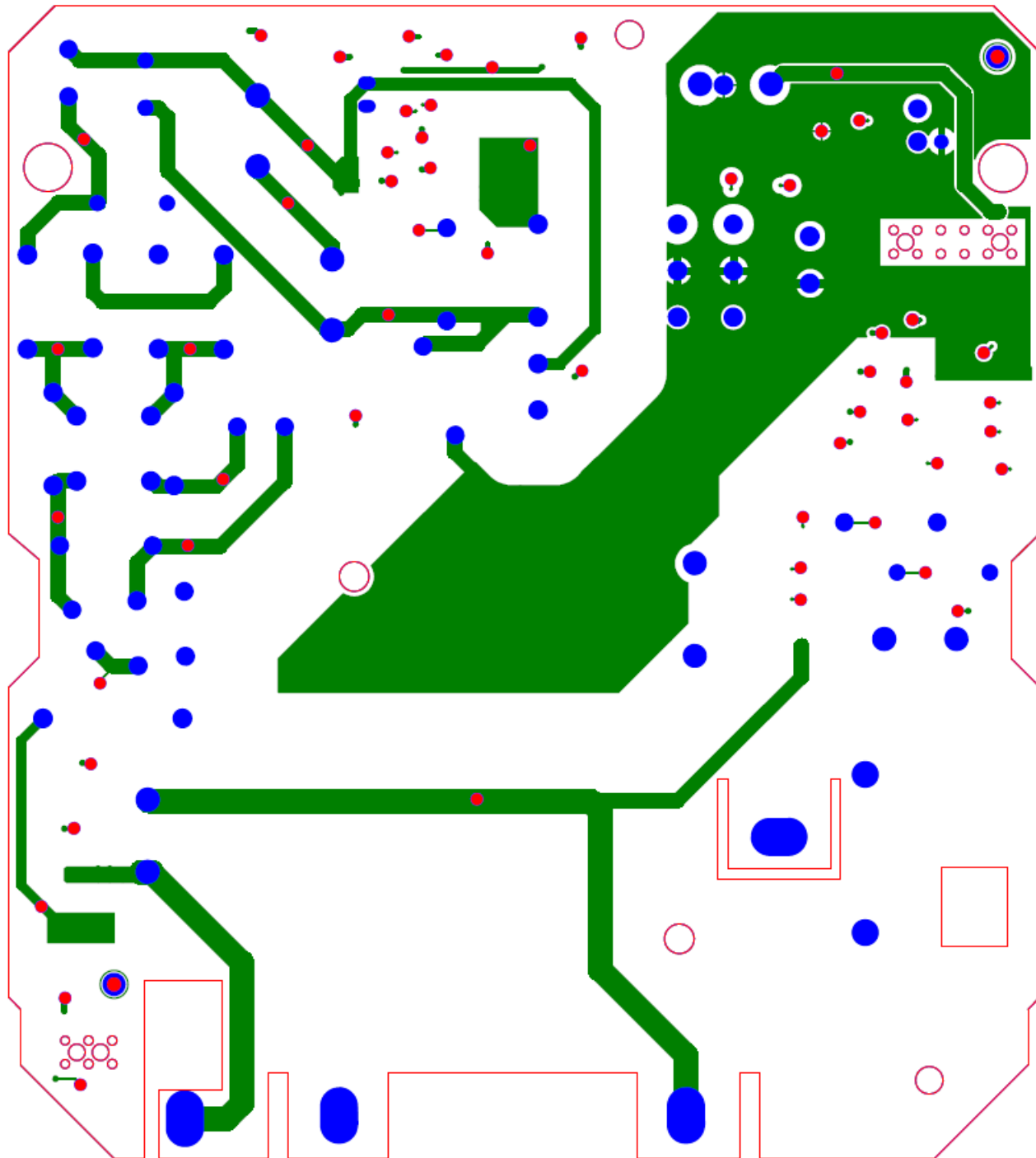
19.1.5 Bottom Silk Screen with component (KF11A019_00 MAN V3.0 20190304)



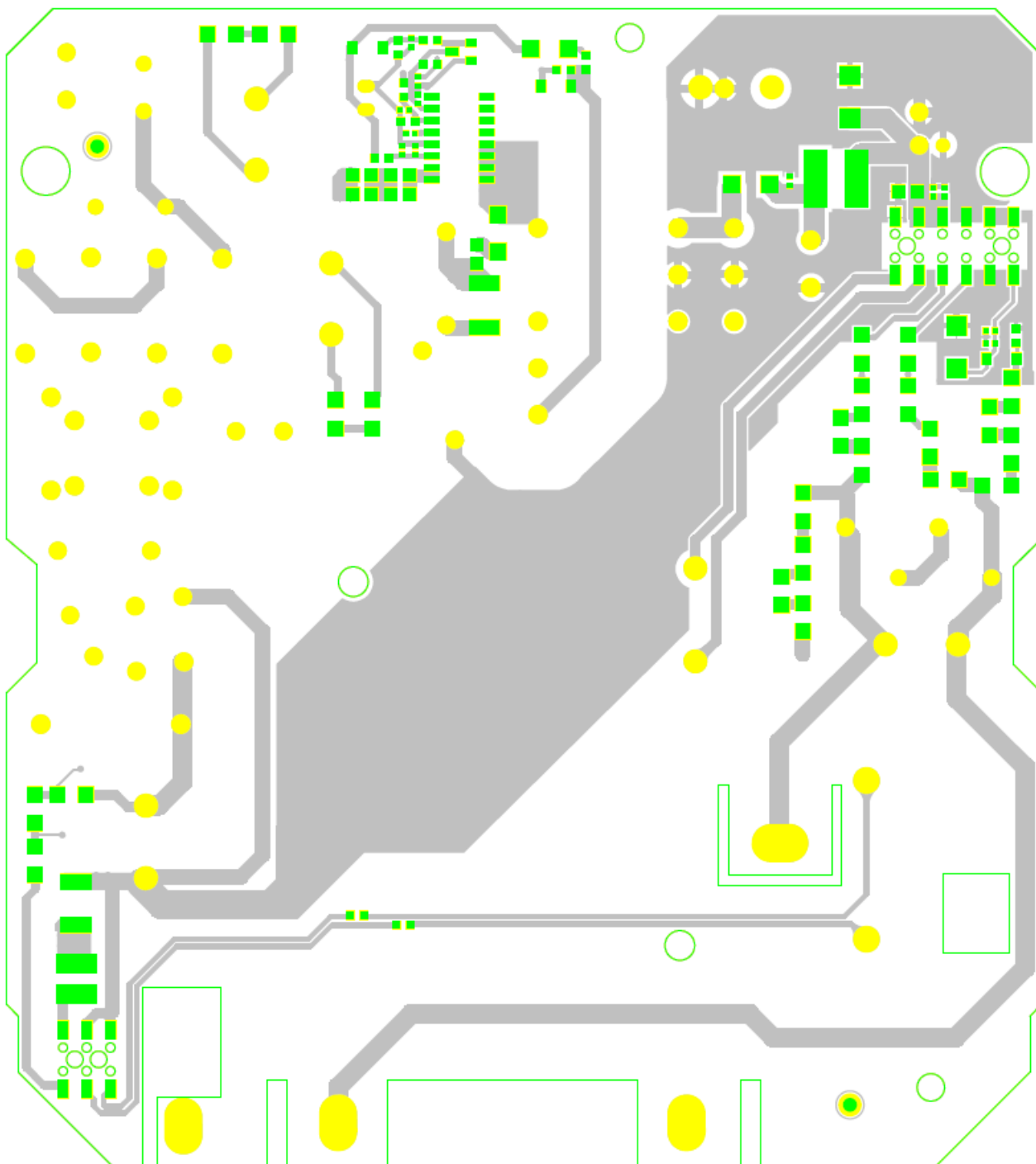
	General Characteristics of Single-Phase Bi-Directional Meter “NEXY-M”	DMI A B 000371	
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19.2 Power Supply PCB (KF11A019_00 POWER V3.0 20190306)

19.2.1 Top layer routing (KF11A019_00 POWER V3.0 20190306)

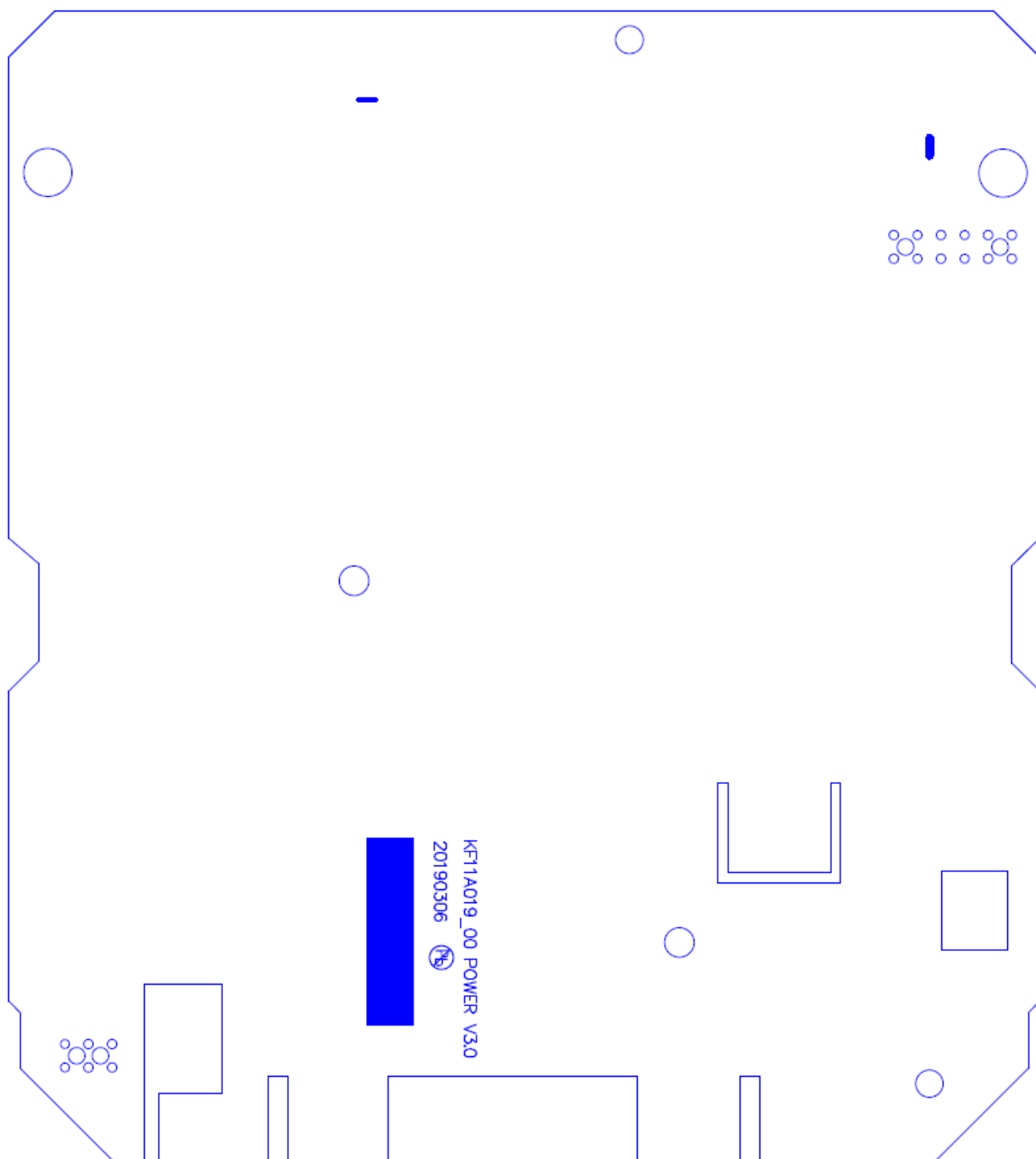


19.2.2 Bottom layer routing (KF11A019_00 POWER V3.0 20190306)

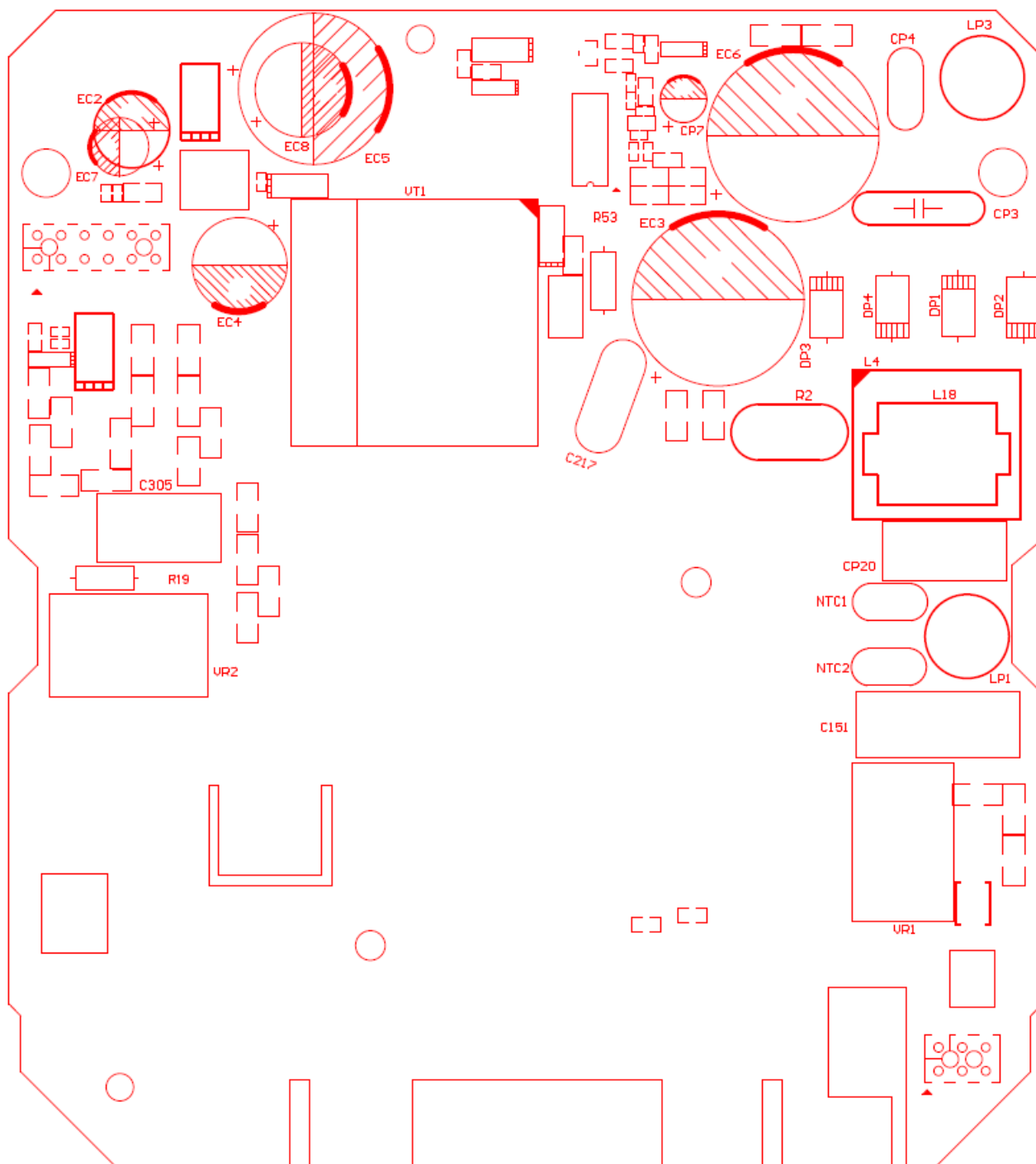


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19.2.3 Top Silk Screen with component (KF11A019_00 POWER V3.0 20190306)



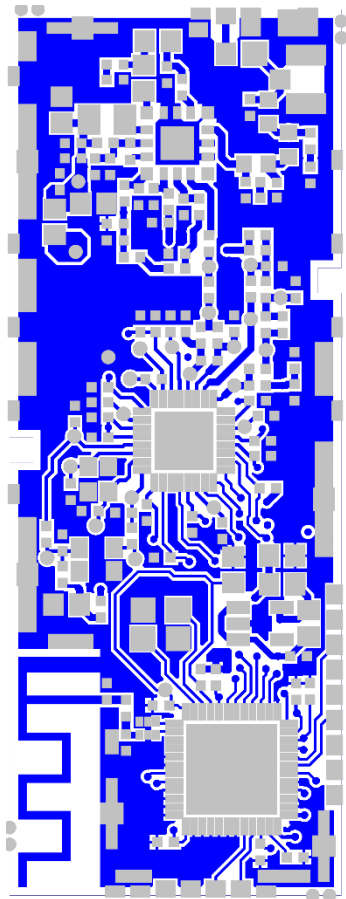
19.2.4 Bottom Silk Screen with component (KF11A019_00 POWER V3.0 20190306)



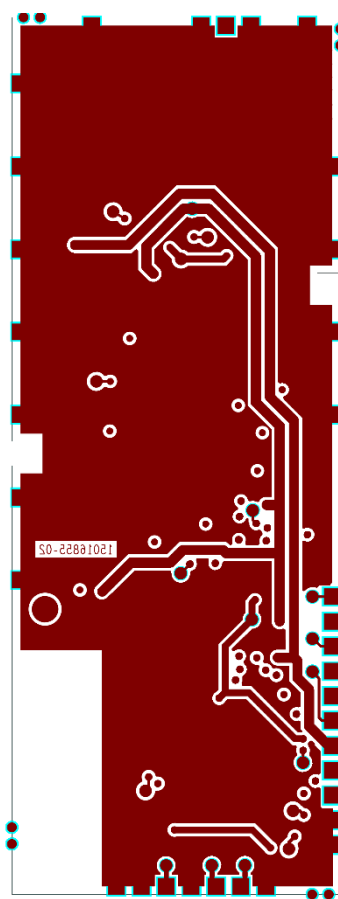
	General Characteristics of Single-Phase Bi-Directional Meter “NEXY-M”	DMI A B 000371
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19.3 Optional RF Module PCB (ID 15016855-02)

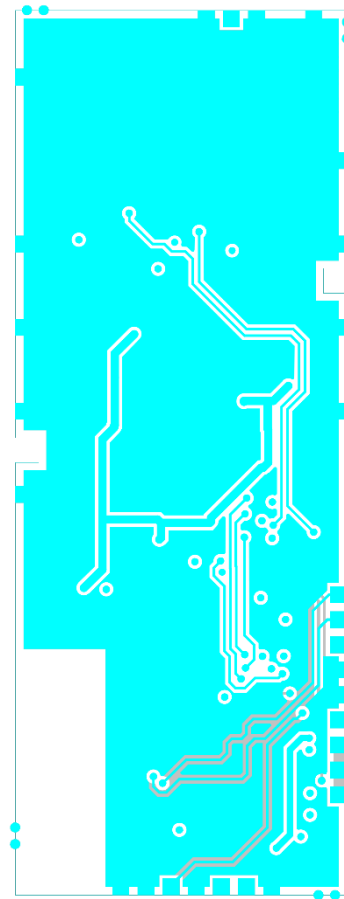
Top layer routing



Bottom layer routing



Inner layer routing



This radio module can be used in conjunction with the antenna “”

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Optional RF Module PCB (ID 15016855-02)

20 NEXY-M Views

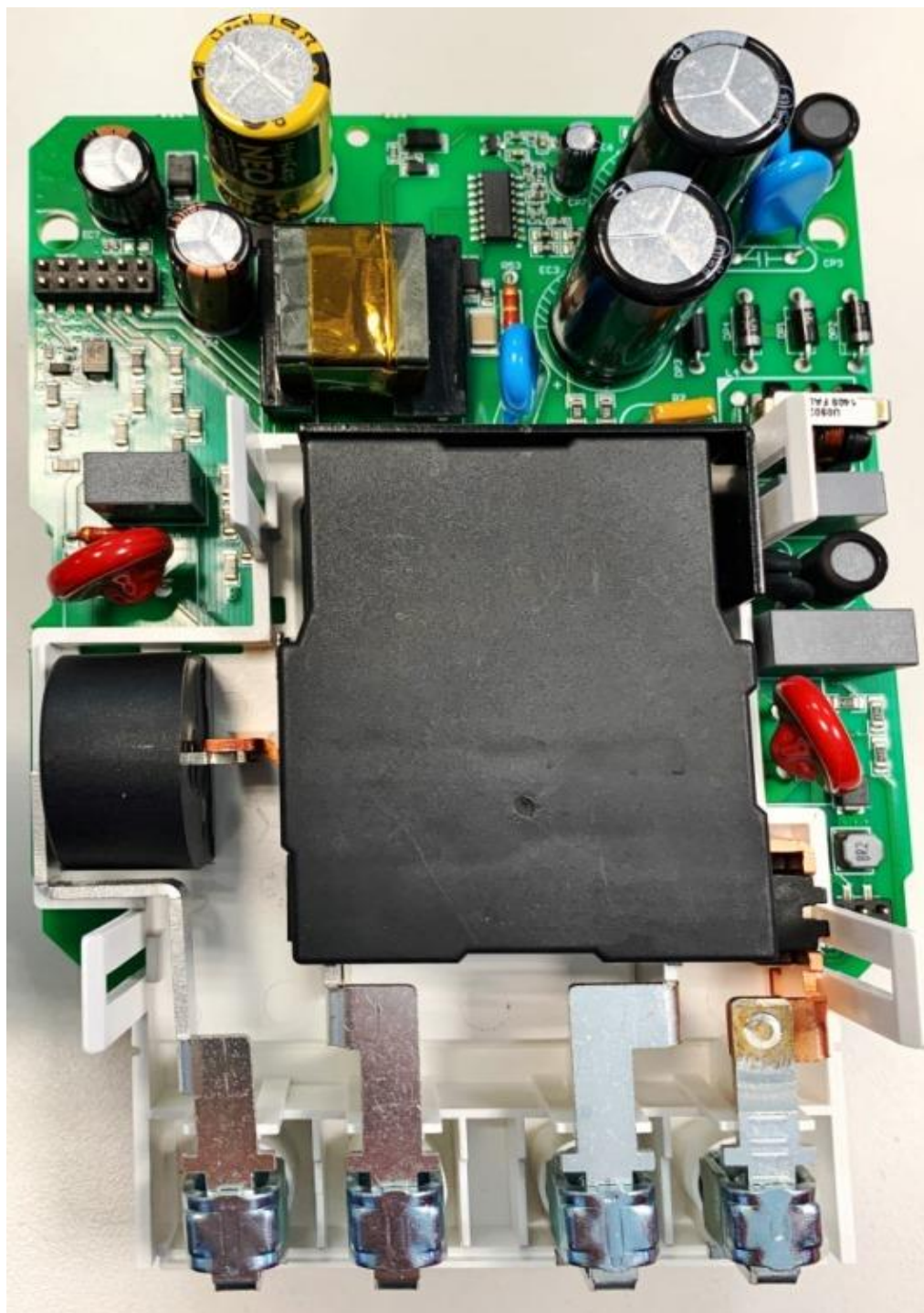
20.1 Top side view



20.2 Top open view



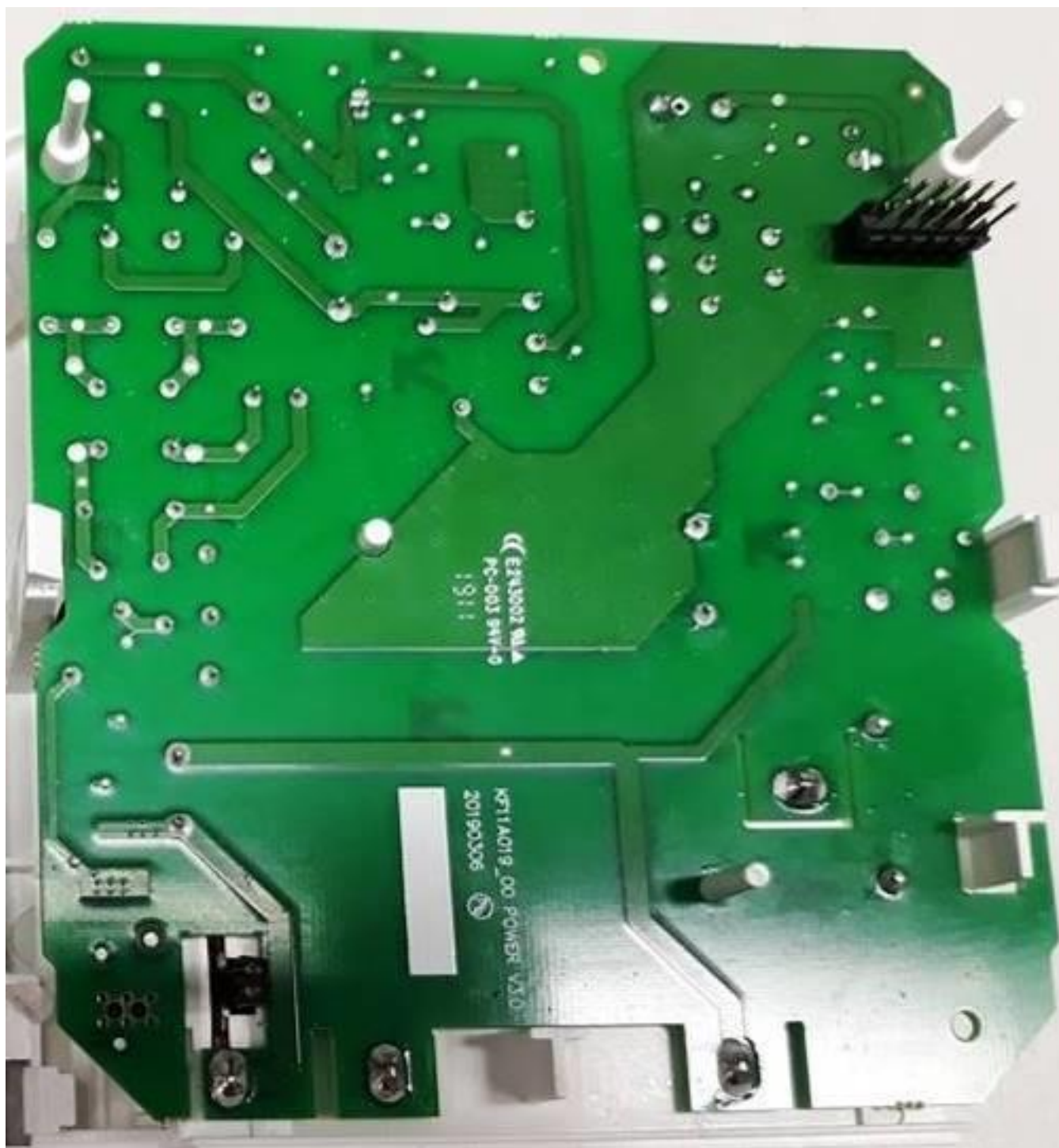
20.3 Bottom open view



20.4 PCBs VIEW



MAIN PCB TOP VIEW



POWER SUPPLY PCB TOP VIEW

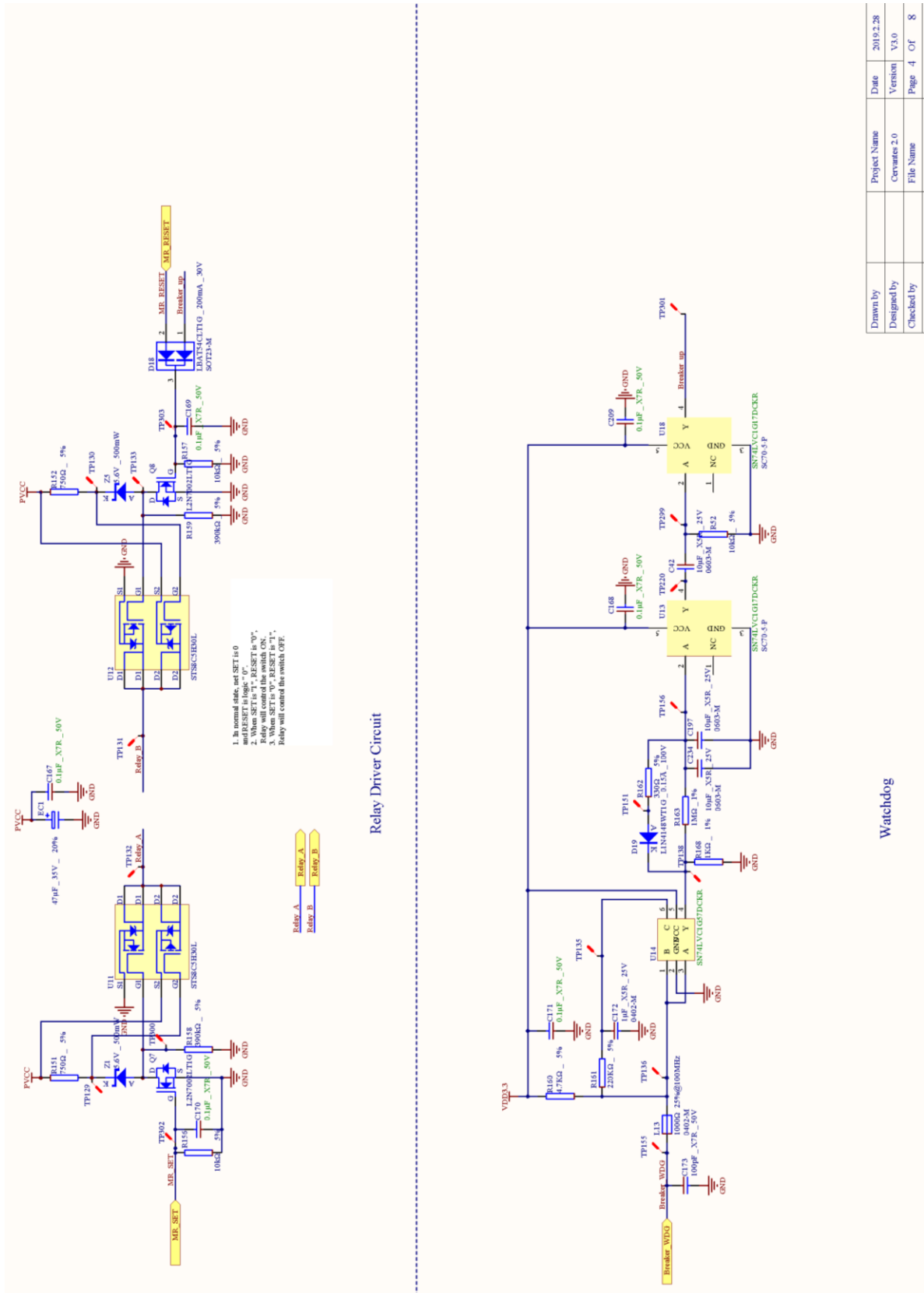


POWER SUPPLY PCB BOTTOM VIEW





Drawn by	Project Name	Date	2019.2.28
Designed by	Cervantes 2.0	Version	V3.0
Checked by	File Name	Page 3	Of 8

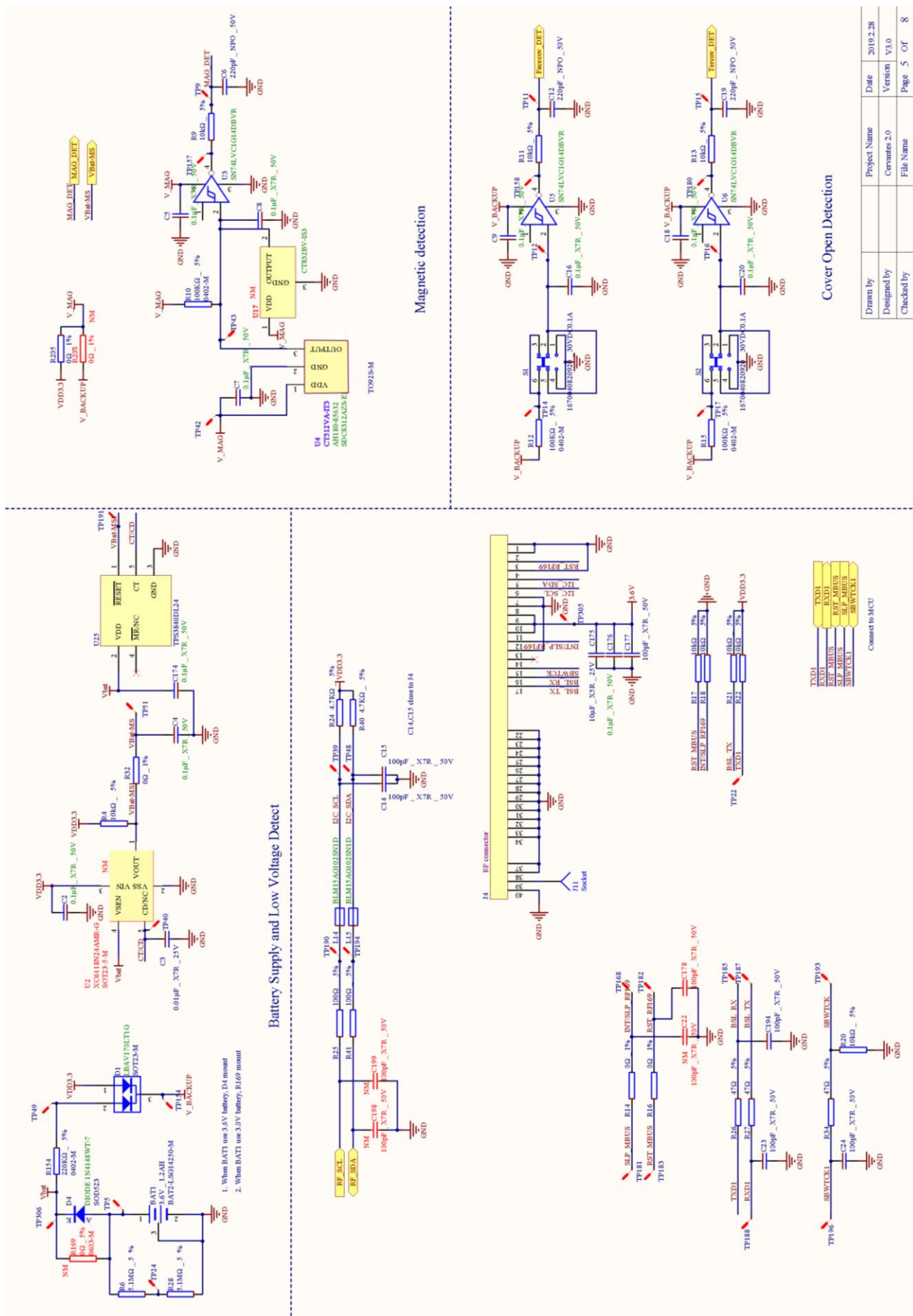


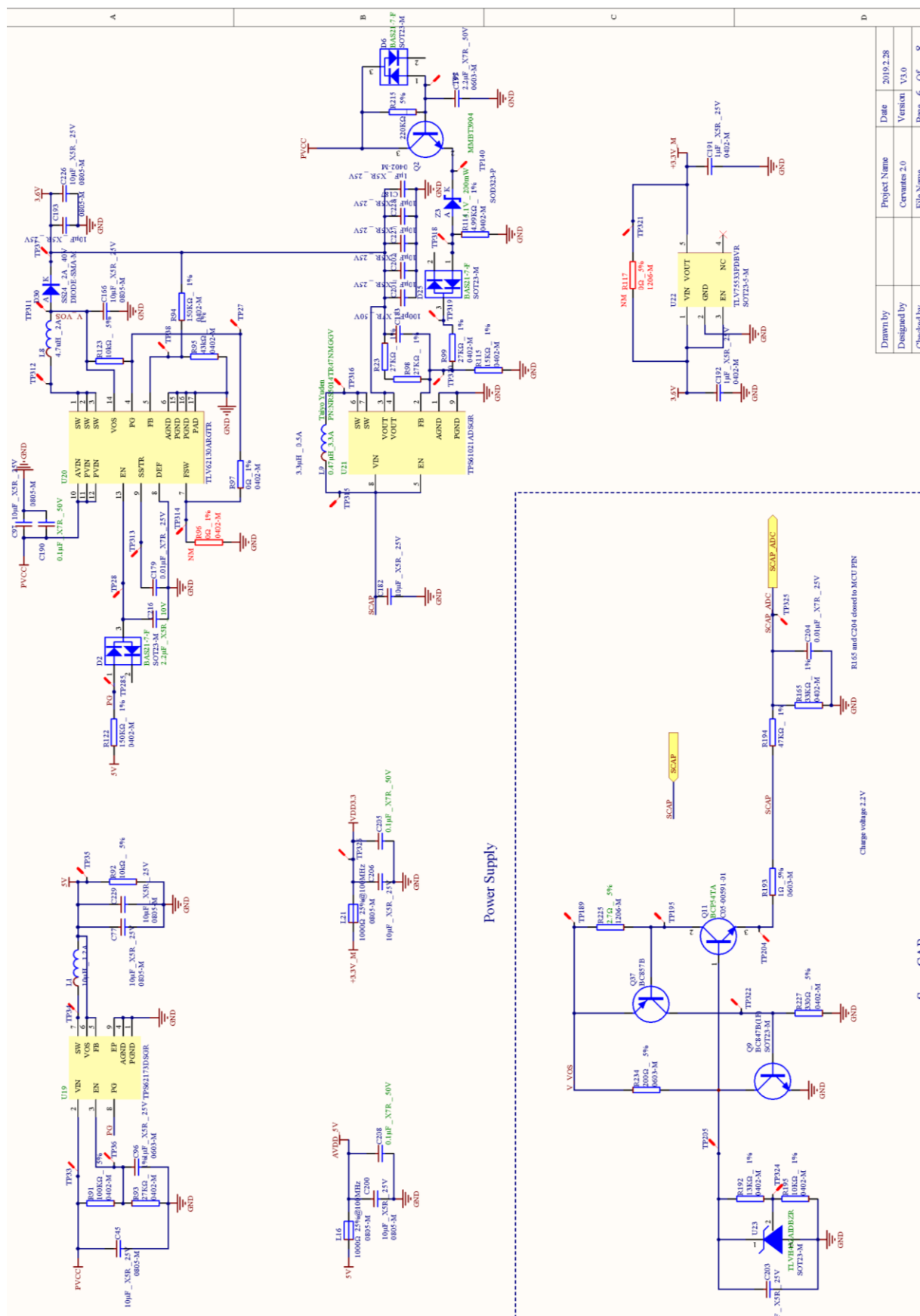
Drawn by	Project Name	Date	2019.2.28
Designed by	Version	Cervantes 2.0	V3.0
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SuperCAP

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Drawn by	Project Name	Date	2019.2.28
Designed by	Curvatures 2.0	Version	V3.0
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Drawn by	Project Name	Date	2019.2.28
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22 Annex B: BOM

22.1 Main PCB

22.1.1 Resistor

#	Schematic Reference	Package	Description	Type
14	R1 R7 R24 R29 R33 R40 R57 R62 R63 R64 R73 R74 R160 R177	0402 SMT Chip	4,7KΩ ±5% 50V 1/16W ±100ppm	N/A
11	R3 R5 R14 R16 R32 R67 R72 R97 R178 R204 R205	0402 SMT Chip	0Ω jumper, 1/16W Thick Film Resistor	N/A
48	R4 R9 R11 R13 R17 R18 R20 R21 R22 R35 R36 R37 R38 R39 R42 R44 R45 R50 R52 R54 R56 R59 R60 R61 R65 R68 R69 R75 R76 R77 R78 R79 R81 R92 R118 R123 R135 R148 R153 R156 R157 R166 R167 R173 R182 R183 R184 R221	0402 SMT Chip	10kΩ ±5% 50V 1/16W ±100ppm	N/A
2	R6 R28	0402 SMT Chip	5,1MΩ ±5% 50V 1/16W ±100ppm	N/A
5	R10 R12 R15 R91 R222	0402 SMT Chip	100KΩ ±5% 50V 1/16W ±100ppm	N/A
4	R23 R93 R98 R99	0402 SMT Chip	27KΩ ±1% 50V 1/16W ±100ppm	N/A
9	R25 R30 R31 R41 R149 R185 R186 R187 R188	0402 SMT Chip	100Ω ±5% 50V 1/16W ±100ppm	N/A
16	R26 R27 R34 R82 R83 R84 R100 R101 R102 R103 R104 R105 R116 R189 R190 R191	0402 SMT Chip	47Ω ±5% 50V 1/16W ±100ppm	N/A
4	R43 R146 R162 R227	0402 SMT Chip	330Ω ±5% 50V 1/16W ±100ppm	N/A
1	R46	0402 SMT Chip	220KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R47 R115	0402 SMT Chip	15KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R48 R49	0603 SMT Chip	0Ω jumper, 1/10W Thick Film Resistor	N/A
1	R51	0402 SMT Chip	430Ω ±5% 50V 1/16W ±100ppm	N/A
8	R71 R110 R111 R131 R138 R147 R168 R176	0402 SMT Chip	1KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R80 R226	0402 SMT Chip	2kΩ ±0.1% 50V 1/16W ±100ppm	REC
1	R87	0402 SMT Chip	62Ω ±5% 50V 1/16W ±100ppm	N/A
2	R88 R89	0603 SMT Chip	120Ω ±1% 50V 1/10W ±100ppm	N/A
4	R113 R119 R120 R125	0603 SMT Chip	1KΩ ±1% 50V 1/10W ±100ppm	N/A
2	R94 R122	0402 SMT Chip	150KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R95	0402 SMT Chip	43KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R109	1206 SMT Chip	330KΩ ±5% 200V 1/4W ±100 ppm Resistor	N/A
1	R112	0603 SMT Chip	10Ω ±5% 50V 1/10W ±100ppm	N/A
1	R114	0402 SMT Chip	4,99KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R121 R124	0603 SMT Chip	3,6Ω ±1% 50V 1/10W ±100ppm	N/A
2	R126 R144	0402 SMT Chip	2KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R127 R145	0402 SMT Chip	6,8KΩ ±1% 50V 1/16W ±100ppm	N/A
10	R128 R129 R132 R133 R136 R137 R140 R141 R150 R194	0402 SMT Chip	47KΩ ±1% 50V 1/16W ±100ppm	N/A
2	R130 R139	2010 SMT Chip	1Ω ±5% 200V 1W ±100ppm Resistor	N/A
2	R142 R143	1206 SMT Chip	100Ω ±1% 200V 1/4W ±100ppm	N/A

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2	R151 R152	0603 SMT Chip	750Ω ±5% 50V 1/10W ±100ppm	N/A
3	R154 R161 R215	0402 SMT Chip	220KΩ ±5% 50V 1/16W ±100ppm	N/A
2	R158 R159	0402 SMT Chip	390kΩ ±5% 50V 1/16W ±100ppm	N/A
1	R163	0402 SMT Chip	1MΩ ±1% 50V 1/16W ±100ppm	N/A
1	R165	0402 SMT Chip	33KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R192	0402 SMT Chip	13KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R193	0603 SMT Chip	1Ω ±5% 50V 1/10W ±200ppm	N/A
1	R195	0402 SMT Chip	10KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R199	0402 SMT Chip	10MΩ ±5% 50V 1/16W ±100ppm	N/A
2	R223 R251	0603 SMT Chip	47kΩ ±0.1% 50V 1/10W ±100ppm	REC
1	R225	1206 SMT Chip	2,7Ω ±5% 200V 1/4W ±100ppm	N/A
3	R232 R241 R250	0603 SMT Chip	1kΩ ±5% 50V 1/10W ±100ppm	N/A
1	R233	0402 SMT Chip	3.3KΩ ±1% 50V 1/16W ±100ppm	N/A
1	R234	0603 SMT Chip	200Ω ±5% 50V 1/10W ±100ppm	N/A
9	RB1 RB2 RB3 RB4 RB5 RB6 RB7 RB8 RB9	0402 SMT Chip array	10KΩ ±5% 50V 1/16W ±200ppm	N/A
1	R8	0402 SMT Chip	NO MOUNT	
4	R66 R70 R134 R155	0402 SMT Chip	NO MOUNT	
1	R58	0402 SMT Chip	NO MOUNT	
3	R96 R181 R206	0402 SMT Chip	NO MOUNT	
1	R117	1206 SMT Chip	NO MOUNT	
1	R169	0603 SMT Chip	NO MOUNT	

22.1.2 Capacitor

#	Schematic Reference	Package	Description	Type
76	C1 C2 C4 C5 C7 C8 C9 C13 C16 C18 C20 C33 C39 C46 C47 C49 C56 C57 C58 C61 C64 C66 C71 C72 C75 C78 C79 C80 C81 C82 C83 C84 C85 C86 C92 C94 C107 C108 C110 C111 C113 C115 C117 C118 C120 C125 C128 C133 C134 C135 C139 C144 C146 C147 C149 C152 C153 C155 C158 C167 C168 C169 C170 C171 C176 C184 C185 C188 C190 C205 C207 C208 C209 C214 C215 C304	0402 SMT Chip	0.1uF ±10% 50V X7R Ceramic Capacitor	N/A
9	C3 C10 C17 C31 C37 C41 C179 C204 C235	0402 SMT Chip	0.01uF ±10% 25V X7R Ceramic Capacitor	N/A
11	C6 C12 C19 C186 C210 C211 C212 C213 C224 C225 C230	0402 SMT Chip	220pF ±5% 50V NP0 Ceramic Capacitor	N/A
12	C11 C29 C30 C32 C34 C35 C145 C154 C172 C187 C191 C192	0402 SMT Chip	1uF ±10% 25V X5R Ceramic Capacitor	N/A
22	C42 C48 C50 C74 C76 C87 C88 C89 C119 C124 C126 C127 C130 C131 C138 C175 C182 C189 C197 C203 C206 C234	0603 SMT Chip	10uF ±20% 25V X5R Ceramic Capacitor	N/A
18	C14 C15 C21 C23 C24 C60 C98 C99 C100 C101 C102 C103 C106 C173 C181 C194 C177 C183	0402 SMT Chip	100pF ±10% 50V X7R Ceramic Capacitor	N/A
4	C25 C26 C140 C141	0402 SMT Chip	22pF ±5% 50V NP0 Ceramic Capacitor	N/A

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8	C38 C40 C122 C123 C129 C132 C136 C137	0402 SMT Chip	33nF ±10% 16V X7R Ceramic Capacitor	N/A
16	C45 C63 C69 C77 C97 C121 C166 C193 C196 C200 C201 C202 C226 C227 C228 C229	0805 SMT Chip	10uF ±10% 25V X5R Ceramic Capacitor	N/A
4	C51 C52 C53 C54	0402 SMT Chip	18pF ±5% 50V NP0 Ceramic Capacitor	N/A
3	C55 C59 C62	0402 SMT Chip	56pF ±5% 50V NP0 Ceramic Capacitor	N/A
12	C27 C28 C73 C90 C91 C93 C95 C104 C105 C112 C114 C116	0402 SMT Chip	47pF ±5% 50V NP0 Ceramic Capacitor	N/A
1	C96	0603 SMT Chip	1uF ±10% 25V X5R Ceramic Capacitor	N/A
2	C109 C307	0402 SMT Chip	10pF ±5% 50V Ceramic Capacitor	N/A
2	C142 C159	0402 SMT Chip	15pF ±5% 50V NP0 Ceramic Capacitor	N/A
2	C143 C160	0402 SMT Chip	4,7pF ±10% 50V NP0 Ceramic Capacitor	N/A
2	C148 C156	1206 SMT Chip	10uF ±10% 25V X5R Ceramic Capacitor	N/A
2	C150 C157	0402 SMT Chip	330pF ±5% 50V NP0 Ceramic Capacitor	N/A
2	C161 C165	0402 SMT Chip	6,8nF ±10% 50V X7R Ceramic Capacitor	N/A
4	C162 C164 C231 C306	0402 SMT Chip	0,01uF ±10% 50V X7R Ceramic Capacitor	N/A
1	C163	0402 SMT Chip	2,2nF ±10% 50V X7R Ceramic Capacitor	N/A
1	C195	0603 SMT Chip	2.2uF ±10% 25V X5R Ceramic Capacitor	N/A
1	C216	0402 SMT Chip	2.2uF ±10% 10V X5R Ceramic Capacitor	N/A
1	EC1	Φ6.3×H11×P2.5	47uF ±20% 35V Electrolytic Capacitor	REC
8	C22 C36 C65 C67 C68 C178 C198 C199	0402 SMT Chip	NO MOUNT	
2	C70 C174	0402 SMT Chip	NO MOUNT	
1	C232	0402 SMT Chip	NO MOUNT	

22.1.3 IC

#	Schematic Reference	Package	Description	Type
1	U1	VSSOP (8)	Dual 2-Input Positive-AND Gate	REC
1	U2	SOT23-5	Voltage Detector low power consumption	TBA
3	U3 U5 U6	SOT-23 (5)	Low-Power Single Schmitt-Trigger Inverter	REC
1	U7	TQFP176 (20 x 20 x 1 mm)	Powerline communication and application system-on-chip, Flash size 1MB	TBA
1	U8	SO8	512-Kbit serial I ² C bus EEPROM	TBA
1	U9	SOP8(208mil)	CMOS 3.0V 8MByte Flash Non-Volatile Memory SPI	TBA
1	U10	QFP28	ADE7953 Measurement IC	TBA
3	U13 U18 U24	SC70 (5)	Single Schmitt-Trigger Buffer	REC
1	U14	SC70 (6)	Low-Power Configurable Multiple-Function Gate	REC
1	U19	WSON(8)	Step-Down Converter I _{max} 0,5A	TBA
1	U20	VQFN (16) 3x3mm	Step-Down Converter I _{max} 3A	TBA
1	U21	WSON (8) 2x2mm	Boost Converter I _{max} 3A	TBA
1	U22	SOT23-5	Low-Dropout Regulator 3,3V	TBA
1	U23	SOT23	Low-Voltage Adjustable Precision Shunt Regulator	REC
1	U30	SOT23-5	Rail-to-rail input/output 8 MHz operational amplifier	TBA
1	U25	SOT23-5	NOT MOUNT	

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22.1.4 Others

#	Schematic Reference	Package	Description	Type
1	BAT1	3PR-7.5	Battery Li, 3.6V 1.2AH	TBA
1	D1	SOT23	BAV170 Double silicon switching diode	REC
3	D2 D6 D25	SOT23	200V 200mA Diode	REC
1	D4	SOD523	Power, Switching Switching Diode 100V 125mA	REC
1	D19	SOD-123	Silicon diode rectifier IF(AV) 150ma VRRM 75v VF(max) 1,2V	REC
1	D5	THT LED 3mm	Red Led	TBA
3	D7 D29 D17	SOT23	BAV99 Double silicon switching diode	REC
4	D8 D9 D13 D14	SMA	Low drop power schottky rectifier IF(AV) 1 A VRRM 30 V VF(max) 0.3 V	REC
2	D10 D12	SMB	Transient voltage suppressor power 600W Vr (typ.) 6.8 V	REC
1	D15	SMB	Transient voltage suppressor power 600W Vr (typ.) 10 V	REC
1	D18	SOT-23	DIODE;SB 30V 200mA	REC
1	D26	THT PHOTOTRANSISTOR. 5mm	Phototransistor	TBA
1	D27	THT LED 5mm	Infrared Led	TBA
1	D30	SMA	Schottky rectifiers IF(AV) 2 A VRRM 40 V VF(max) 0.85	REC
1	J1	SMT Connector 1x12 way	SMT connector 1x12 way	REC
1	J14	SMD PIN3x2, pitch 2,54mm	Female Connector 3 x 2 ways, Bottom Entry	REC
1	J15	SMD PIN6x2, pitch 2,54mm	Female Connector 6 x 2 ways, Bottom Entry	REC
1	L1	SMT 4 x 4 x 1,8mm	10uH ±20% Rated current 1,2A Inductor	REC
1	L5	0603 SMT Chip	Multilayer Ferrite 330Ω ±25% @100MHz, Rated current = 2A, Rdc =0.15Ω	REC
2	L6 L7	0603 SMT Chip	Multilayer Ferrite 120Ω ±25% @100MHz Rdc=0.15Ω 600mA -55°C to+125°C	REC
1	L8	SMT 6 x 6 x 2,8 mm	4,7μH ±20% 3,7A Inductor	REC
1	L9	SMT 4,9 x 4,9 x 1,4 mm	0,47uH ±30% Rated current 3,3A Inductor	REC
1	L11	SMT 7,4 x 7,3 x 3,5mm	15uH ±20% 2A Inductor	REC
1	L12	SMT 7,8 x 7 x 5mm	470uH ±10% 0,34A Inductor	REC
5	L2 L3 L13 L14 L15	0402 SMT Chip	Multilayer Ferrite 1000Ω ±25% @100MHz Rdc=0,7Ω 300mA -55°C to+125°C	REC
3	L16 L21 L40	0805 SMT Chip	Multilayer Ferrite 1000Ω ±25% @100MHz Rdc=0,12Ω 1500mA -55°C to+125°C	REC
2	L38 L39	L-CD43-M	Multilayer Ceramic Inductor High Frequency, 220nH ±3%, 0.1A	REC
3	Q1 Q3 Q9	SOT23	45V, 225mW NPN Small Signal Transistor	REC
1	Q2	SOT23	40V NPN Small Signal Transistor	REC
2	Q4 Q5	SOT23-6L	N-channel 30 V, 0.021 Ω typ., 6 A	REC
2	Q7 Q8	SOT23	MosFET N-channel 60 V, Rds(on) max <13,5Ω (@10V) Id =0,3 A	REC

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1	Q11	SOT-223	BCP54 NPN Transistor	REC
1	Q37	SOT23	45V, 225mW PNP Small Signal Transistor	REC
2	S1 S2	6DIP 7×7×16mm	Push Switch 30VDC 0,1A	TBA
1	S3	4SMT 6.2×6.2×5mm	Push Switch 6VDC 50mA	TBA
1	T1	15.4x13.97x13.2mm SMD	PLC COUPLING TRANSFORMER 1:1 1mH ±30% @100kHz	TBA
1	U4	SIP-3	High sensitivity micropower omnipolar Hall-effect switch	TBA
2	U11 U12	SOP8	N-channel 30 V, 0.018 Ω typ., 8 A, P-channel 30 V, 0.045 Ω typ., 5 A Power MOS	REC
1	Y1	3,2 x 2,5 mm SMT Chip 4PAD	24.000MHz Crystal ±10ppm CL= 12pF Temperature range= -40+85°C	TBA
1	Y2	3.2 x 1.5 mm SMT Chip 2PAD	32.768KHz Crystal ±10ppm CL= 12,5pF ESR 60kohm max	TBA
1	Y3	SMT HC-49	3.579545MHz ±20ppm 16pF -40°C to +85°C ±30ppm	TBA
2	Z1 Z5	SOD-123	5,6V 500mW Zener Diode	REC
1	Z3	SOD-323	5,1V 200mW Zener Diode	REC
1	J3	SIP 1,27mm 2x4 - SMD-M	NO MOUNT	
1	U17	SOT23	NO MOUNT	
1	PIN HEADER	PIN 3X2 pitch 2.54mm, h 21mm	PIN HEADER 3x2 WAY gold plated (see Drawing)	REC
1	PIN HEADER	PIN 6X2 pitch 2.54mm, h 21mm	PIN HEADER 6x2 WAY gold plated (see Drawing)	REC
1	CURRENT TRANSFORMER 60A*	w=15mm, h=26,5mm, THT 2pins at distance 17mm	60A current transformer PCB mount	TBA
1	CURRENT TRANSFORMER 100A*	w=22mm, h=29,5mm, THT 2pins at distance 17mm	100A current transformer PCB mount	TBA
1	Liquid Crystal Display	DISPLAY MODULE	Custom COG module	TBA
1	Antenna	Customized antenna assembly	169MHz stick antenna	TBA
1	RADIO MODULE		Custom Radio Module 169MHz + BLE Board	TBA

*Only one Current transformer will be installed, depending on the I_{max} of the NEXY-M.

22.2 Power Supply PCB

22.2.1 Resistor

#	Schematic Reference	Package	Description	Type
1	R19	15,2mm THT AXI	12KΩ ±5% 350V 1W ±250ppm Resistor	N/A
1	R53	15,2mm THT AXI	220KΩ ±5% 350V 1W ±250ppm Resistor	N/A
4	R55 R85 R86 R90	0805 SMT Chip	1Ω ±1% 200V 1/4W ±200 ppm Chip Resistor	REC
3	R106 R107 R108	1206 SMT Chip	330KΩ ±5% 200V 1/4W ±100 ppm Resistor	N/A
1	R164	0603 SMT Chip	1kΩ ±5% 50V 1/10W ±100ppm	N/A
1	R174	0402 SMT Chip	10kΩ ±5% 50V 1/16W ±100ppm	N/A
1	R196	0805 SMT Chip	100Ω ±5% 150V 1/8W ±100ppm	N/A

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1	R197	0603 SMT Chip	27K Ω $\pm$ 1% 50V 1/10W $\pm$ 100ppm	N/A
1	R198	0402 SMT Chip	1.3K Ω $\pm$ 1% 50V 1/16W $\pm$ 100ppm	N/A
4	R207 R209 R210 R212	1206 SMT Chip	3,3M Ω $\pm$ 5% 200V 1/4W $\pm$ 100ppm	N/A
1	R211	0603 SMT Chip	10 Ω $\pm$ 5% 50V 1/10W $\pm$ 100ppm	N/A
1	R213	0603 SMT Chip	1.2k Ω $\pm$ 1% 50V 1/10W $\pm$ 100ppm	N/A
1	R214	0402 SMT Chip	8,2K Ω $\pm$ 1% 50V 1/16W $\pm$ 100ppm	N/A
1	R224	0402 SMT Chip	1M Ω $\pm$ 5% 50V 1/16W $\pm$ 100ppm	N/A
8	R228 R229 R230 R231 R242 R243 R244 R245	1206 SMT Chip	1M Ω $\pm$ 1% 200V 1/4W $\pm$ 100ppm	N/A
1	R235	0603 SMT Chip	33K Ω $\pm$ 5% 50V 1/10W $\pm$ 100ppm	N/A
8	R236 R238 R239 R240 R246 R247 R248 R249	1206 SMT Chip	200k Ω $\pm$ 5% 200V 1/4W $\pm$ 100ppm	N/A
1	R252	0603 SMT Chip	33 Ω $\pm$ 5% 50V 1/10W $\pm$ 100ppm	N/A
1	R253	0603 SMT Chip	2.7k Ω $\pm$ 1% 50V 1/10W $\pm$ 100 ppm	N/A
1	R208	0805 SMT Chip	NO MOUNT	

22.2.2 Capacitor

#	Schematic Reference	Package	Description	Type
1	C43	1812 SMT Chip	4.7nF $\pm$ 10% 1000V X7R Ceramic Capacitor	REC
1	C44	0402 SMT Chip	1uF $\pm$ 10% 25V X5R Ceramic Capacitor	N/A
1	C151	W:17,5mm H:13,5mm T:7,5mm pitch 15mm	0.15uF $\pm$ 10% 305VAC X2 Capacitor THB 1000h	REC
1	C180	0603 SMT Chip	330nF $\pm$ 10% 16V X7R	N/A
1	C217	Φ 11.5 x P10	2200pF $\pm$ 20% 400Vac Y5V -25°C/+85°C Ceramic Capacitor	REC
3	C218 C219 C220	0402 SMT Chip	0.1uF $\pm$ 10% 50V X7R Ceramic Capacitor	N/A
1	C221	0603 SMT Chip	0.47uF $\pm$ 10% 16V X7R Ceramic Capacitor	N/A
2	C222 C308	0402 SMT Chip	1000pF $\pm$ 10% 50V X7R Ceramic Capacitor	N/A
1	C223	0402 SMT Chip	4700pF $\pm$ 10% 50V X7R Ceramic Capacitor	N/A
1	C233	0402 SMT Chip	1uF $\pm$ 10% 10V X5R Ceramic Capacitor	N/A
1	C305	W13xH13xT7, pitch 10mm	15nF $\pm$ 10% 440VAC X1 Capacitor	REC
1	CP3	Φ 14xP7.5	10nF -20%~+80% 2KV Y5V ceramic capacitor	REC
1	CP4	Φ 14xP5	4,7nF -20%~+80% 2KV Y5V ceramic capacitor	REC
1	CP7	Φ 5 x H11 x P2	10uF $\pm$ 20% 50V 105°C Electrolytic Capacitor	REC
1	CP20	W13xH12xT6, pitch 10mm	10nF $\pm$ 10% 440VAC X1 Capacitor	REC
1	EC2	Φ 8xH12xP3.5	270uF $\pm$ 20% 25V Electrolytic Capacitor	REC
2	EC3 EC6	Φ 16 x H30 x P7.5	68uF $\pm$ 20% 400V -40/+105°C Electrolytic Capacitor	REC
1	EC4	Φ 10 x H20 x P5	1000uF $\pm$ 20% 25V -40/+105°C Electrolytic Capacitor Low ESR	REC
1	EC5	Φ 16xH25xP7.5	SUPER CAP 25F -10% ~ +30% 2.7/3.0V	TBA
1	EC7	Φ 6.3 x H11 x P2.5	NO MOUNT	
1	EC8	Φ 10xH30xP5	NO MOUNT	

22.2.3 Others

#	Schematic Reference	Package	Description	Type
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1	D3	SMA	Power Schottky rectifier IF(AV) 1A, VRRM 150V, VF(max) 0.67V	REC
1	D11	SMB	Transient voltage suppressor power 600W Vr (typ.) 18V	REC
1	D16	SOD123	Silicon diode rectifier IF(AV) 200mA VRRM 75V VF(max) 1.25V	REC
1	D20	SOD123	Low capacitance Schottky diodes 100V,200mA	REC
1	D24	SMA	Ultrafast high voltage VRRM 800 V IF(AV) 1 A	REC
4	DP1 DP2 DP3 DP4	DO-15	High Voltage Rectifier IF(AV)=0.2A VRRM=5000V	REC
1	DP7	SMB	Power Schottky rectifier IF(AV) 3A, VRRM 100V, VF(max) 0.68V	REC
1	J12	SMD PIN6x2, pitch 2,54mm	Female Connector 6 x 2 ways, Bottom Entry	REC
1	J16	SMD PIN3x2, pitch 2,54mm	Female Connector 3 x 2 ways, Bottom Entry	REC
1	L10	SMT 7.3mm x 7.3mm x 4.5mm	8.2uH ±20% 2.7A Inductor	REC
2	L17 L20	0603 SMT Chip	Multilayer Ferrite 1000Ω ±25% @100MHz Rdc=0.15Ω 600mA - 55°C to +125°C	REC
1	L18	8x7mm pitch	Common choke >18mH Rdc<8ohm Idc<0.1A	REC
1	L19	SMT 6 x 6 x 2,8 mm	4,7μH ±20% 3,7A Inductor	REC
1	LP1	Φ9.0 × H12 × P5.0 THT	1mH ±10% Ir = 510mA Inductor	REC
1	LP3	Φ9.0 × H12.2 × P5.0 THT	1.5mH ±10% RDC<=3.8 ohm, Imax 0.43A Inductor	REC
2	NTC1 NTC2	Φ8.5×P5	NTC Thermistor 4.7Ω Imax 3.0 A	REC
1	Q6	SOT23	45V, 225mW NPN Small Signal Transistor	REC
1	R2	pitch 5mm/ 9.5mm Dia Radial Disc	PTC Thermistor 10Ω Itrip 0.3A	REC
1	UP1	SO16N-M	Resonant switching power supply controller	TBA
2	VR1 VR2	17x13.5x6.5 mm	460VAC/610VDC Thermal protected varistors	TBA
1	VT1	EP17 8pin	EP-17 switching transformer N1:N2:N3:N4=70:10:13:70	TBA
1	Z2	SMB	DIODE;TVS 6.40~7.07V 600W	REC
1	Z4	SMB	15V 1W zener diode	REC
1	Z6	SOD123-M	ZENER DIODE 19V 500mW	REC
1	L4	13x10mm pitch, W:18mm, H:17.5mm, T:16mm	NO MOUNT	

22.3 Safety critical components

Please refer to the following attached file:

Section 22.3 - Safety critical components NEXY-M (2021.04.xx)

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COMPONENTS TYPES	
TBA:	TBA defined components require formal approval by e-distribuzione upon presentation of samples and data sheets by the assembler. These components can only be used after the approval is given.
REC:	REC defined components require formal approval by e-distribuzione upon presentation of documentation (supplier reference, data sheet and part number) and - only if requested - of samples by the assembler. These components can only be used after the approval is given.
N/A:	N/A defined components require only a formal information to e-distribuzione (supplier reference, data sheet and part number) by the assembler prior to purchase order issue to the supplier. The assembler will guarantee that the components used comply, as a minimum, with the approved suppliers requirements and components datasheet.